

THE M-CATS

FALL 2026

Reproduction

BIOLOGY CHAPTER 2

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The M-CATS at the University of Kentucky

2.1

Chromosomes

Organisms have a different amount of genetic information (as DNA)

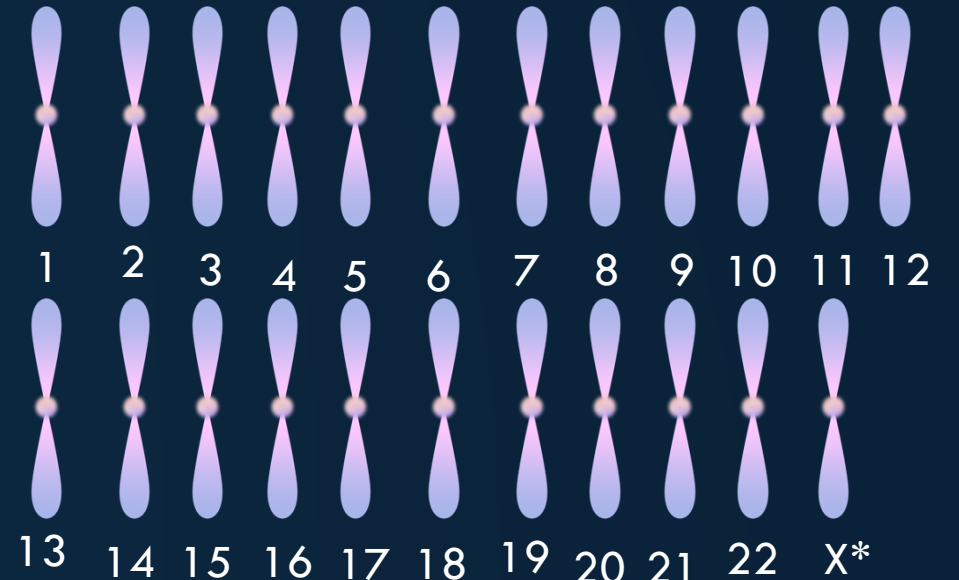
Organized into linear strands called
Chromatids



Single Chromatid

Humans code our entire
genome in

23 UNIQUE chromatids



Organisms have a different amount of genetic information (as DNA)

Organized into linear strands called
Chromatids



Single Chromatid

Aliens codes their entire
genome in

2 UNIQUE chromatids





1



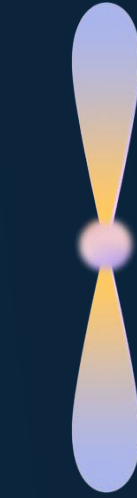
2

Duplicate

Cause
y'know
Just in case



1



Copy
1



2



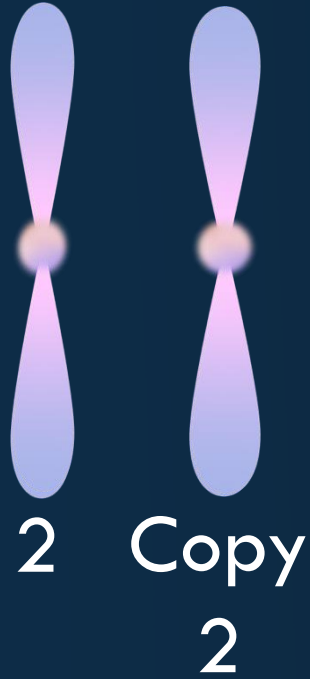
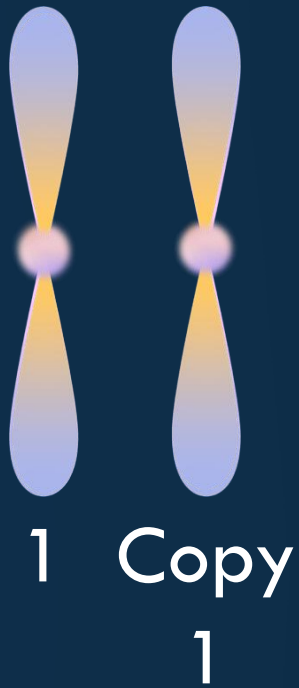
Copy
2

One copy =
Haploid (n)

n= 1

Two copies =
Diploid (2n)

2n= 2



THE RULE

We NEED all of our cells
to be diploid and the
right number of
chromosomes

All cells then need how many copies
of chromosomes?

2 copies!

Diploid!! Remember the rule!

THE RULE

We NEED all of our cells to be diploid and the right number of chromosomes



Reproduction



2 IDENTICAL daughter
cells

The Mother Cell

THE RULE

We NEED all of our cells to be diploid and the right number of chromosomes



The Mother Cell

2 IDENTICAL daughter
cells

2.2

Mitosis

Mitosis

A form of cellular reproduction which produces two identical diploid daughter cells

This is embedded in the cell cycle:

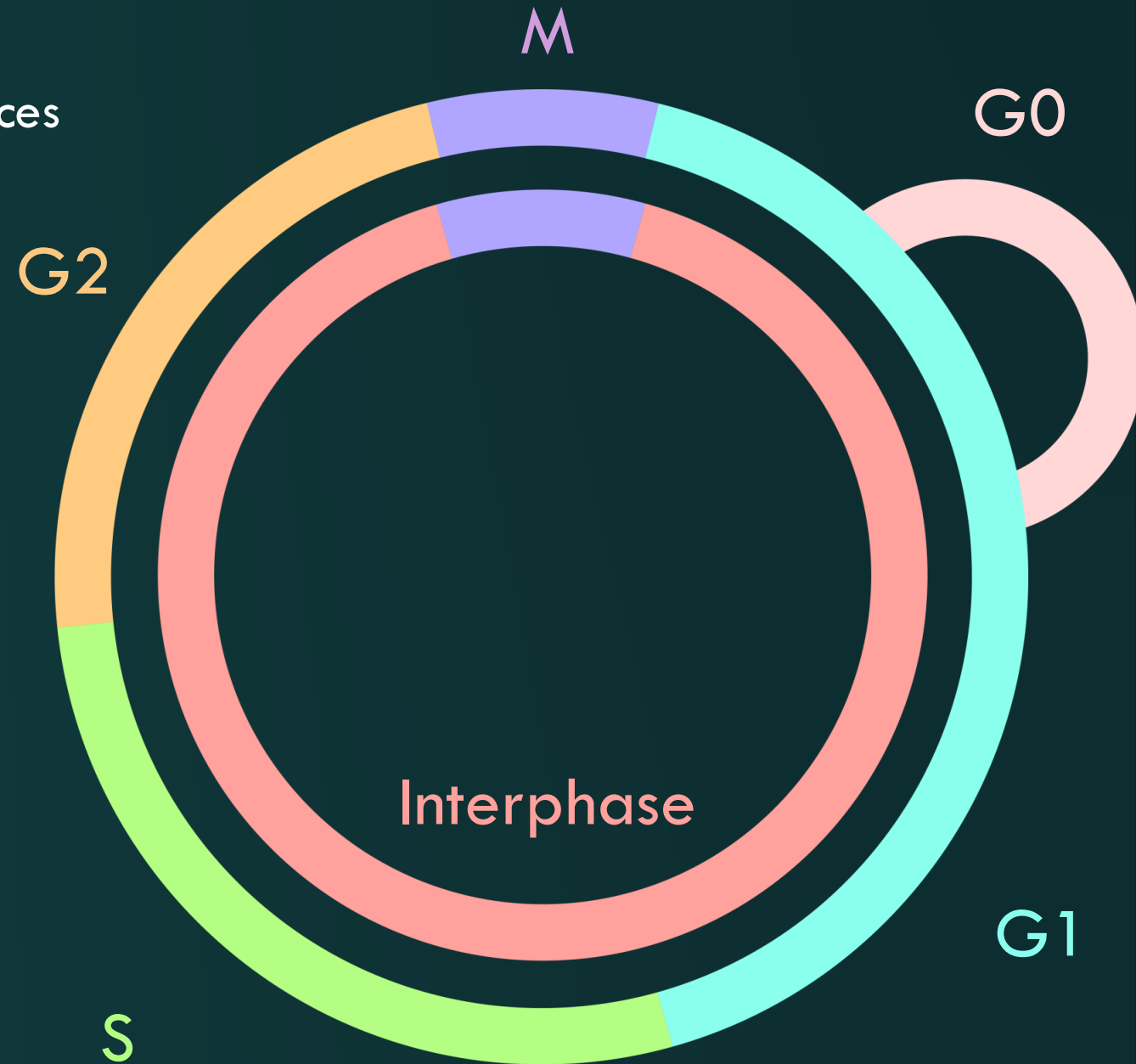
This is similar to getting ready for a Child.

Do you have enough money?

Enough food?

Enough rooms?

You need to check if you have the proper resources. And acquire them.

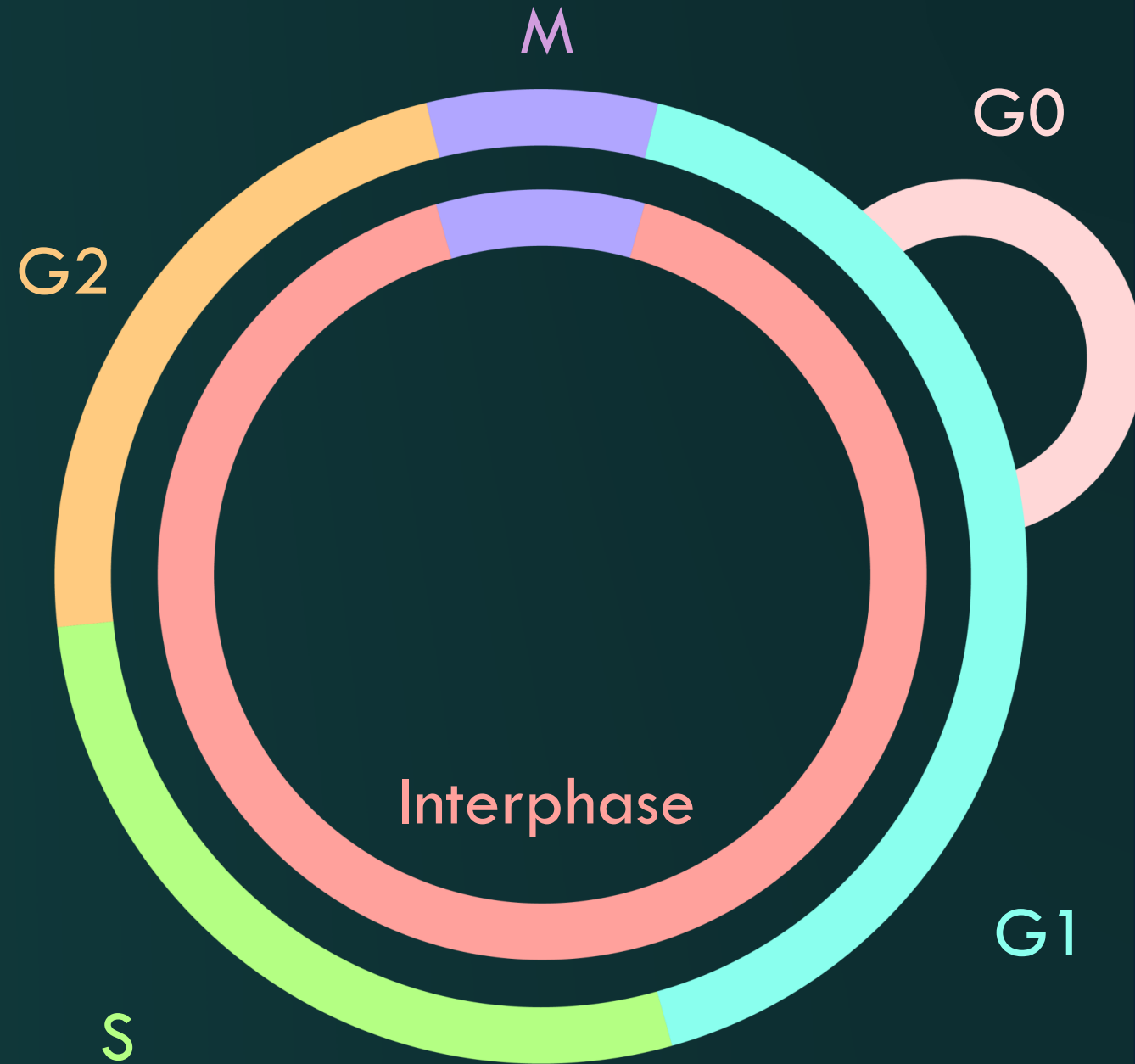


G0

You are either cycling (replicating,
Which also involves getting ready to
replicate)
Or you are chilling

Neurons

Cardiomyocytes



G1

Pre-synthetic growth phase 1

Increases synthesis of organelles and size, so it has enough for both daughter cells and so it can support the cell as it undergoes replication

Restriction Point

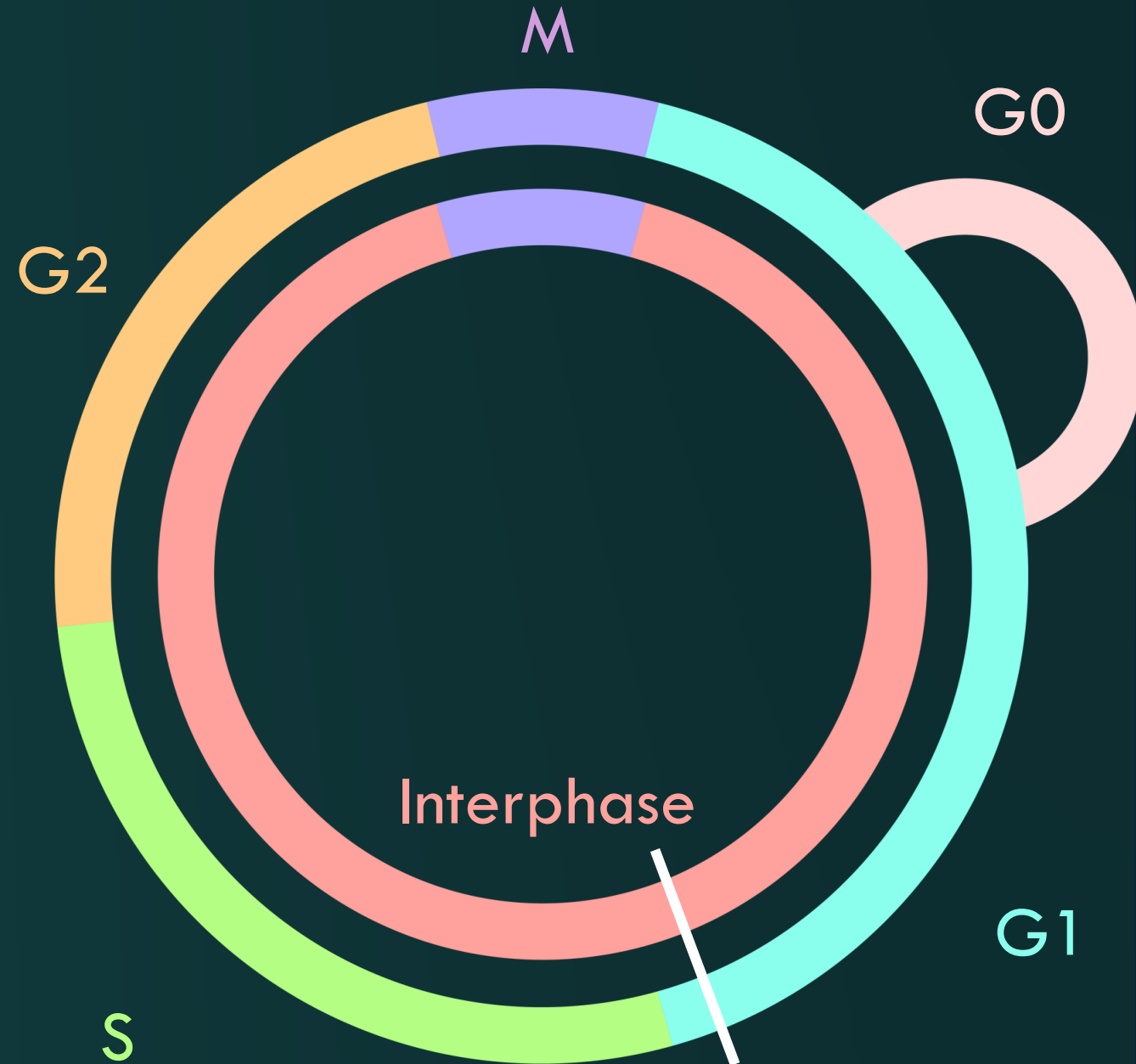
Checks if the cell is adequate enough to replicate

Is my DNA damaged?

Do I have nutrients?

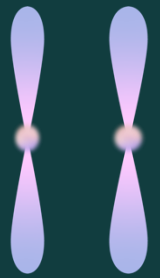
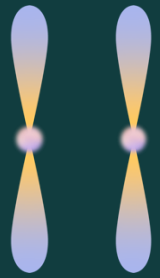
Did I grow enough?

Did I make enough organelles?



S

Replicate all of the DNA

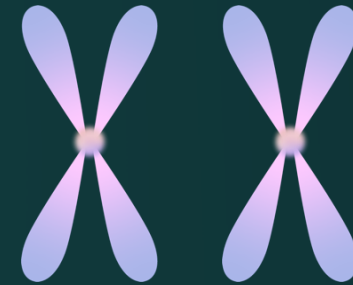
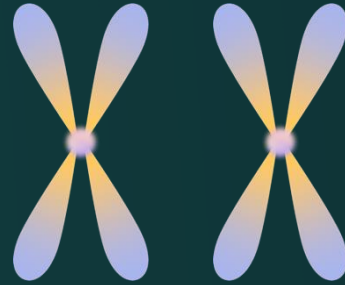


$2n$

4 chromosomes

The Mother Cell

S-phase



$2n$

4 chromosomes
8 sister chromatids

Post S-phase

G2

M

Interphase

S

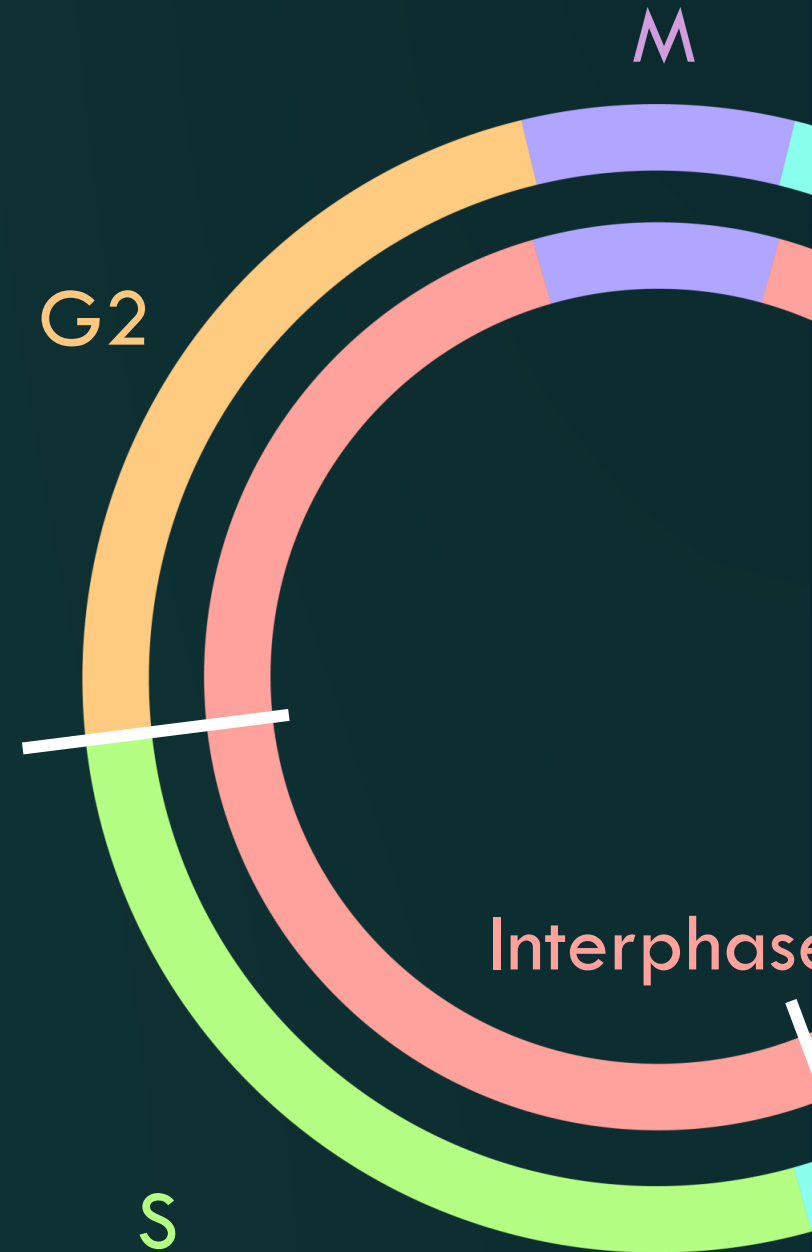
G2

Before G2 we have

S/G2 checkpoint

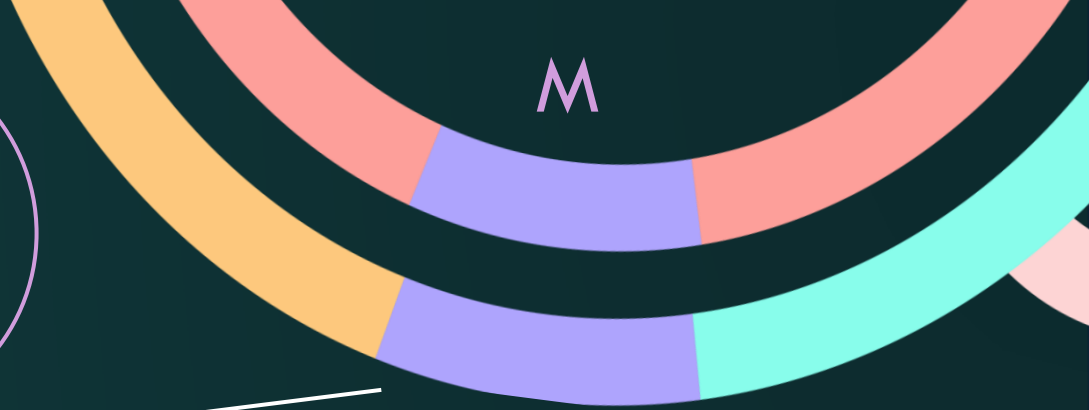
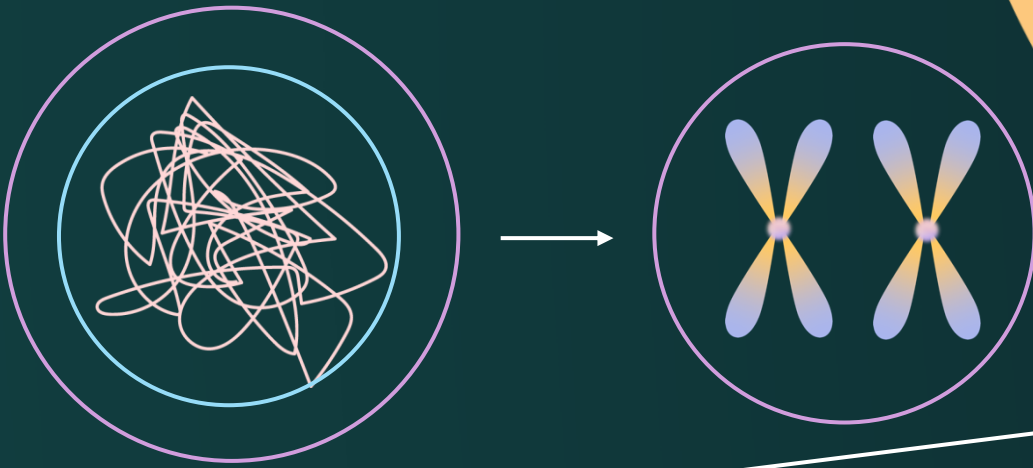
We check if the DNA was replicated properly

Final growth and checks



M

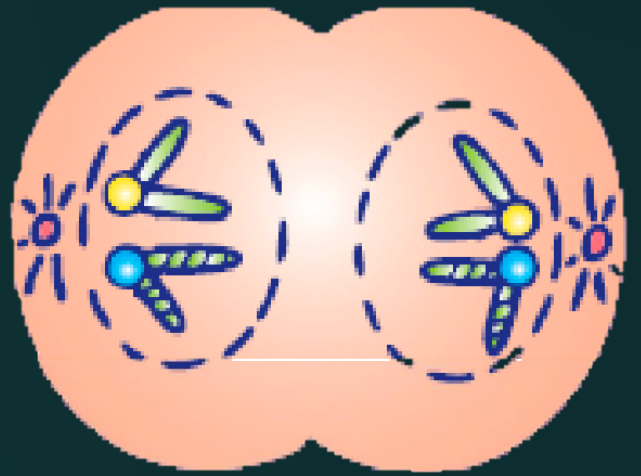
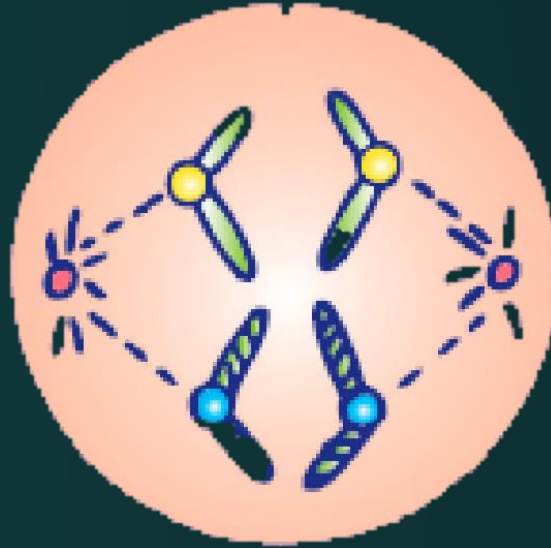
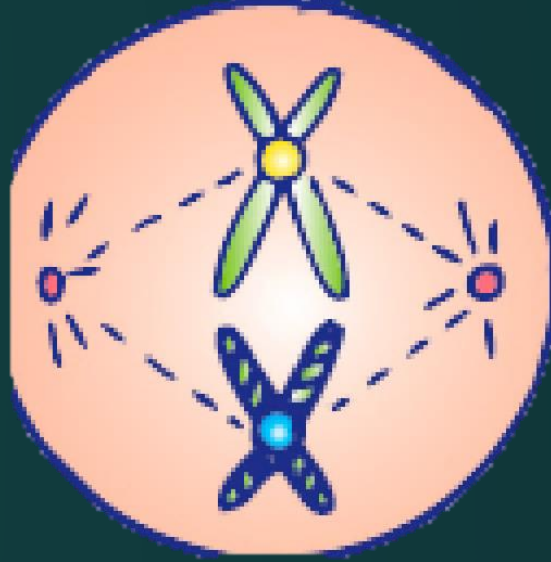
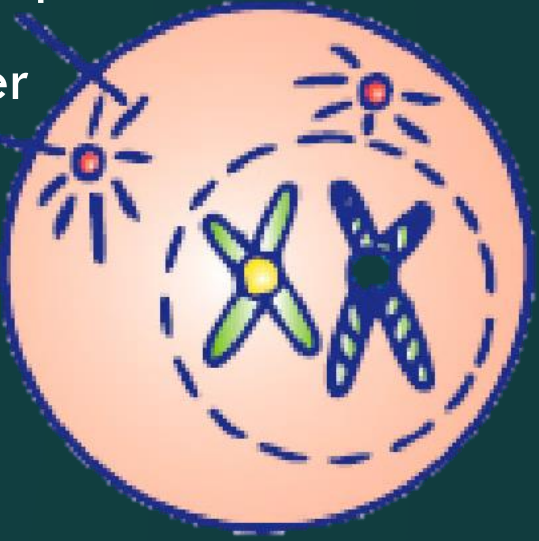
4 phases



Metaphase plate

spindle

aster



Prophase

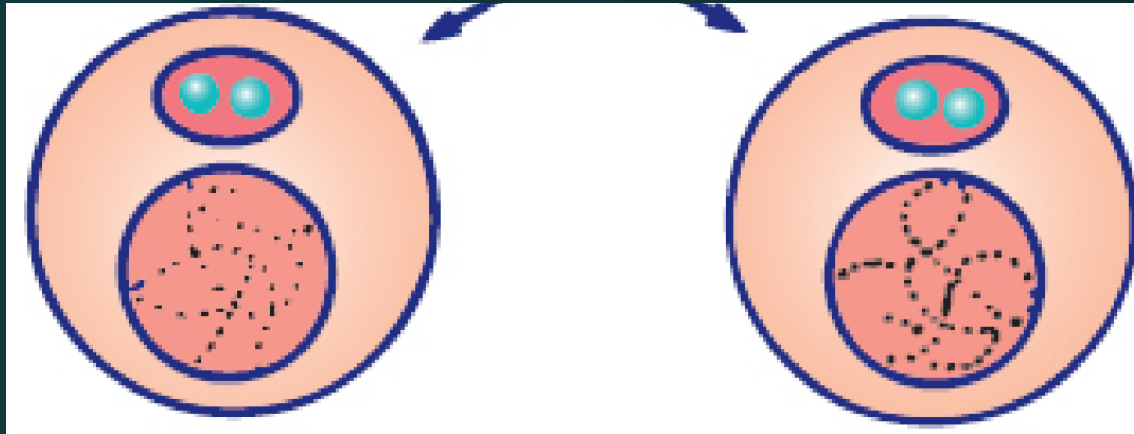
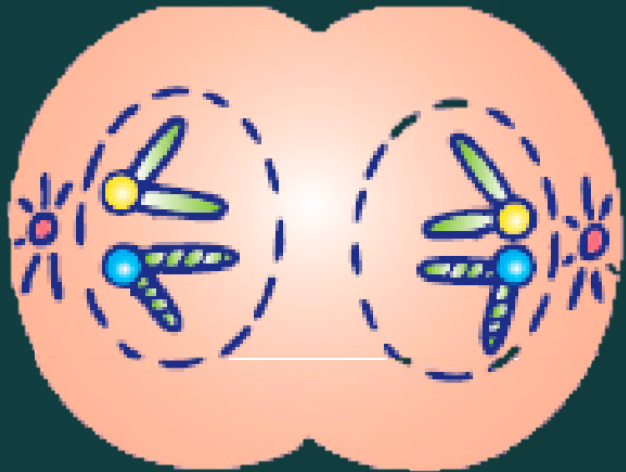
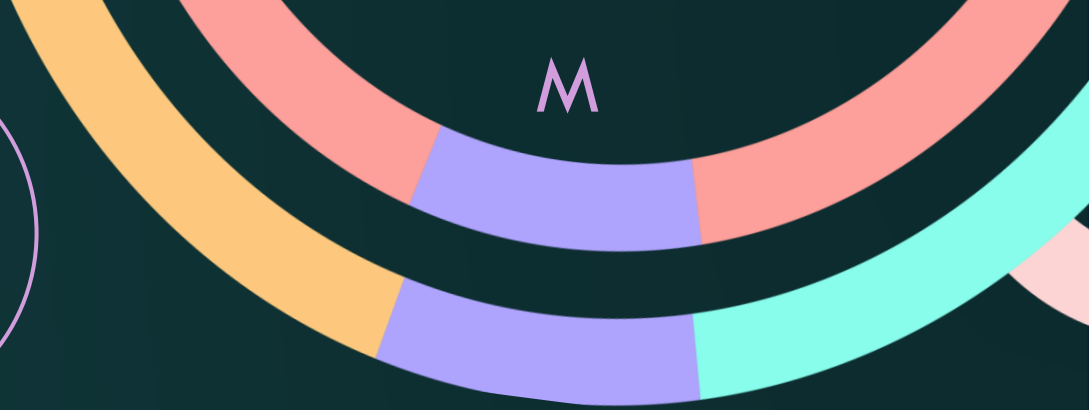
Metaphase

Anaphase

Telophase

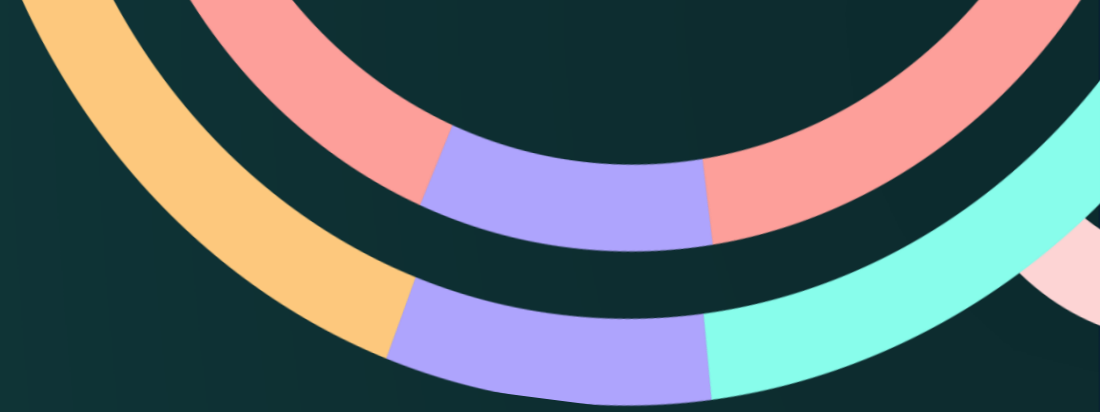
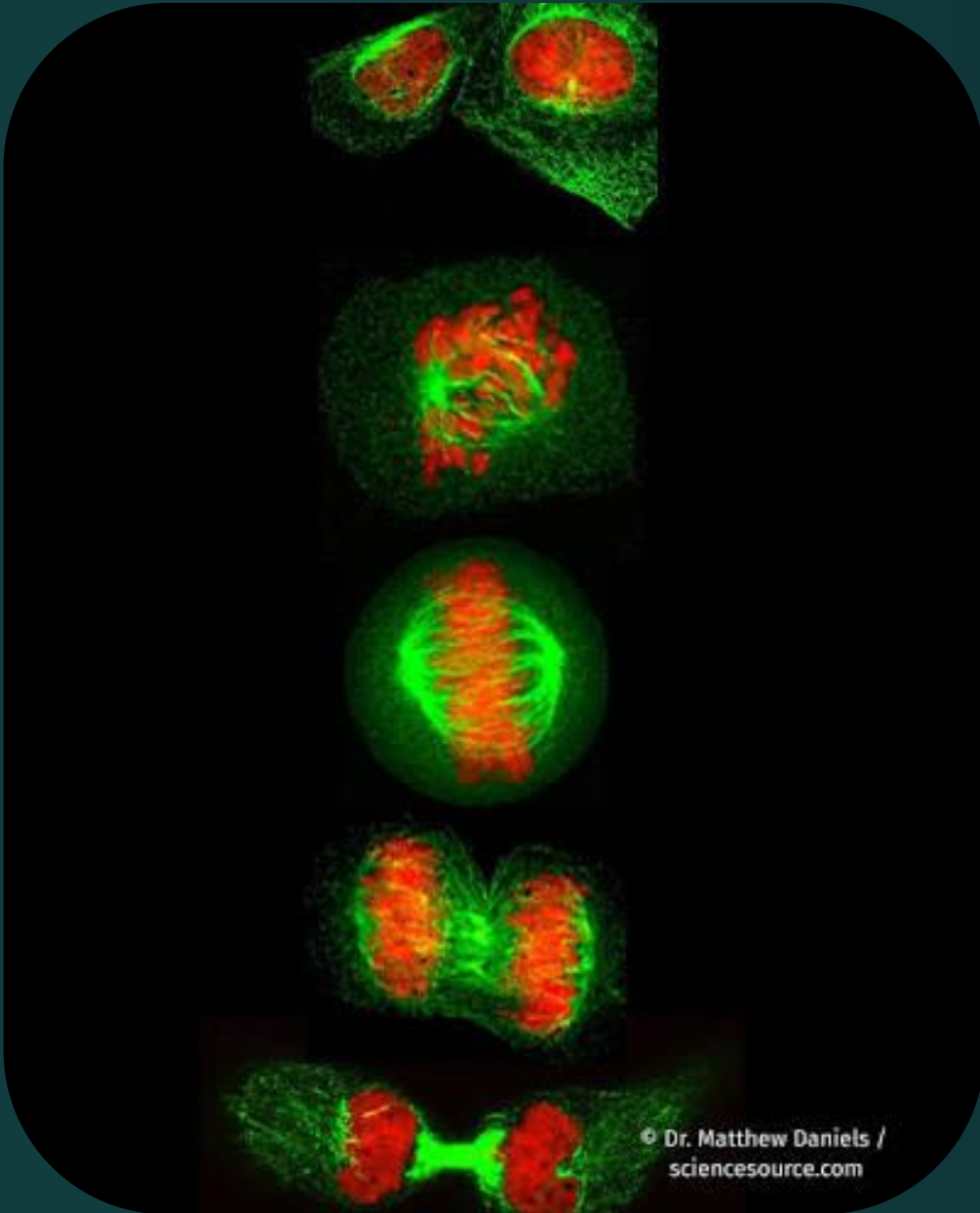
M

4 phases

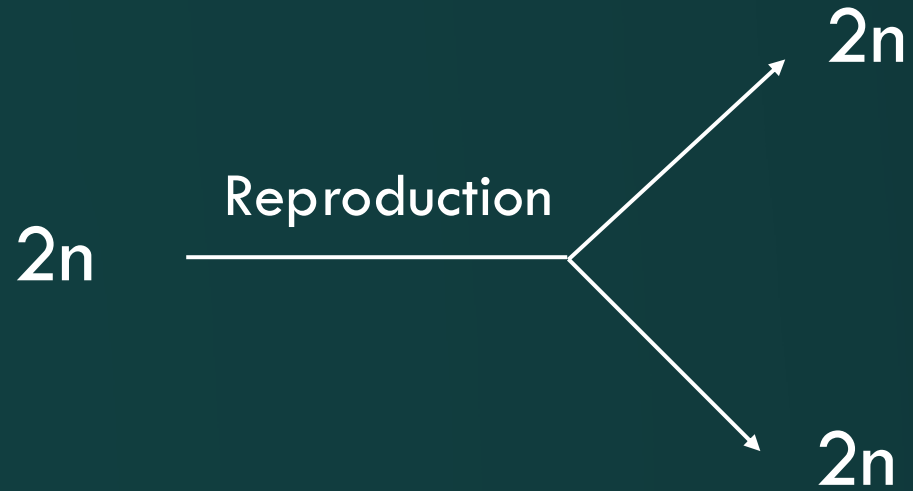


Telophase

Cytokinesis



Mitosis



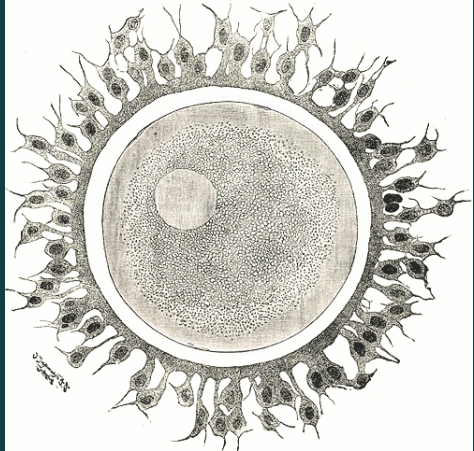
In humans every cell should have

23 chromosomes

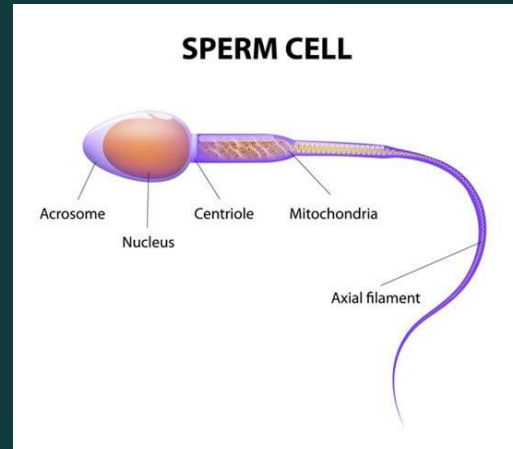
2 copies of each

46 chromatids

$2n$ 23



$2n$ 23



In humans every cell should have

Diploid 23 chromosomes

2 copies of each

46 chromatids

So how can we
prevent this?

we need a method
where the sex cells

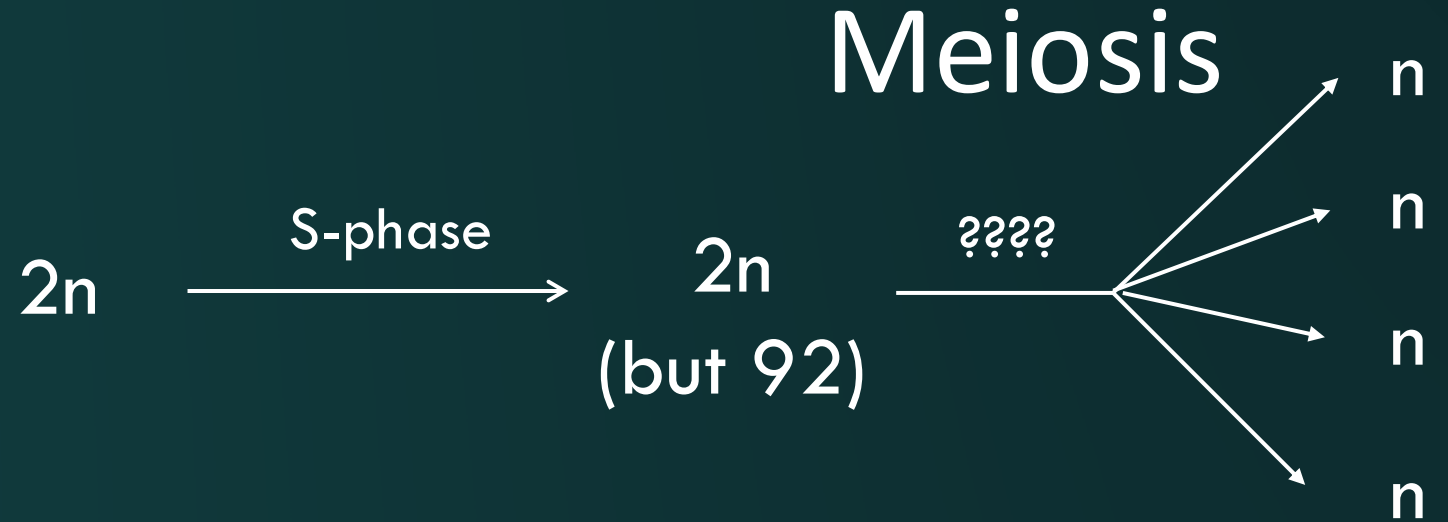
Can be haploid



$4n$ 46 chromosomes
92 chromatids

So how can we
prevent this?

we need a method
where the sex cells
Can be haploid



if haploid we just need 23 in each.

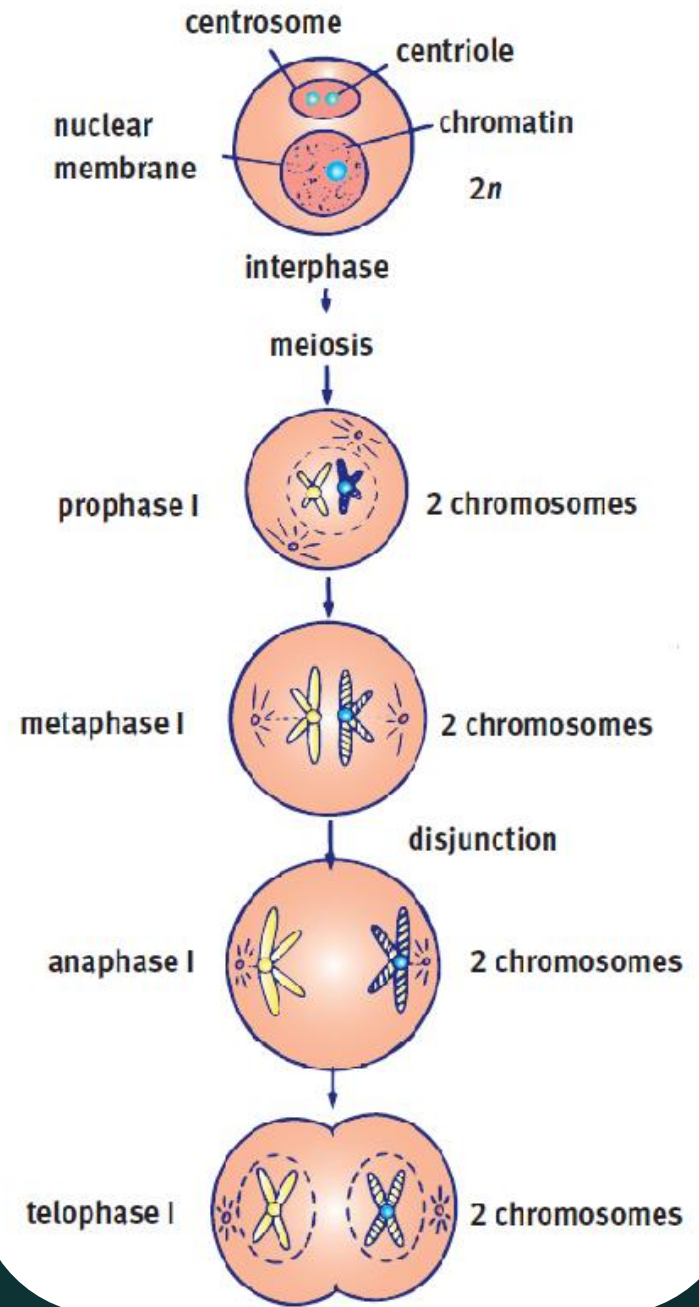
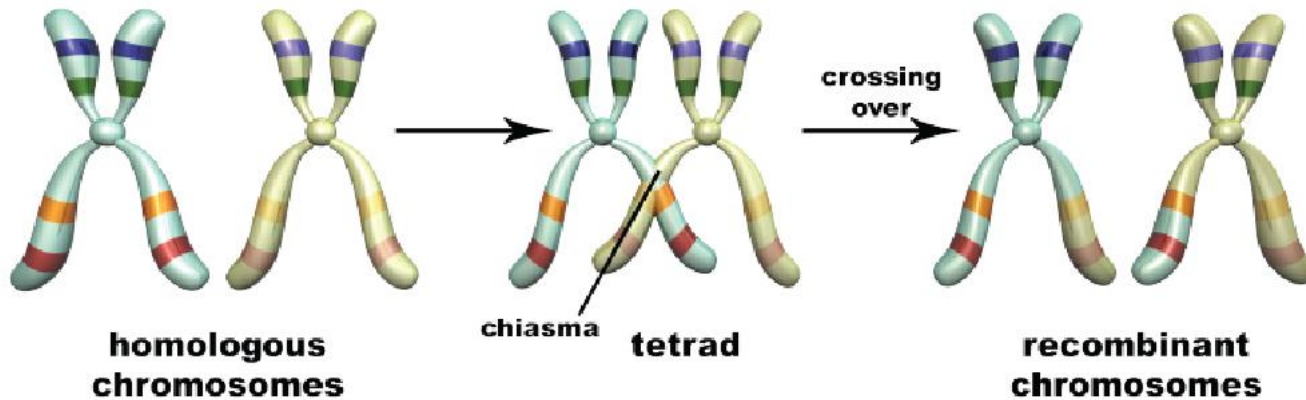
We start with 92.

$92 / 23 = 4$. Can make 4 haploid cells

Meiosis

Main differences during stage 1

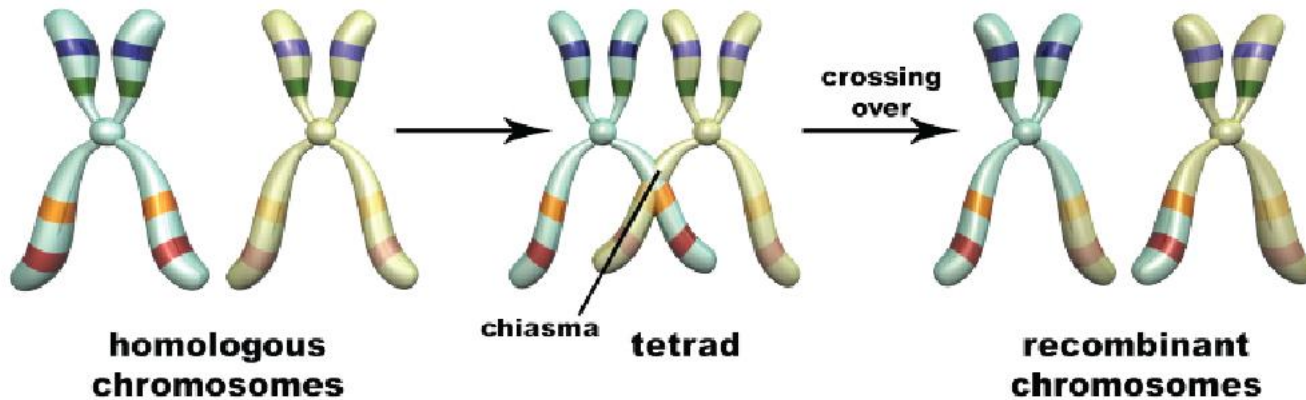
1. We pull apart homologous chromosomes, not sister chromatids
2. Crossing over occurs



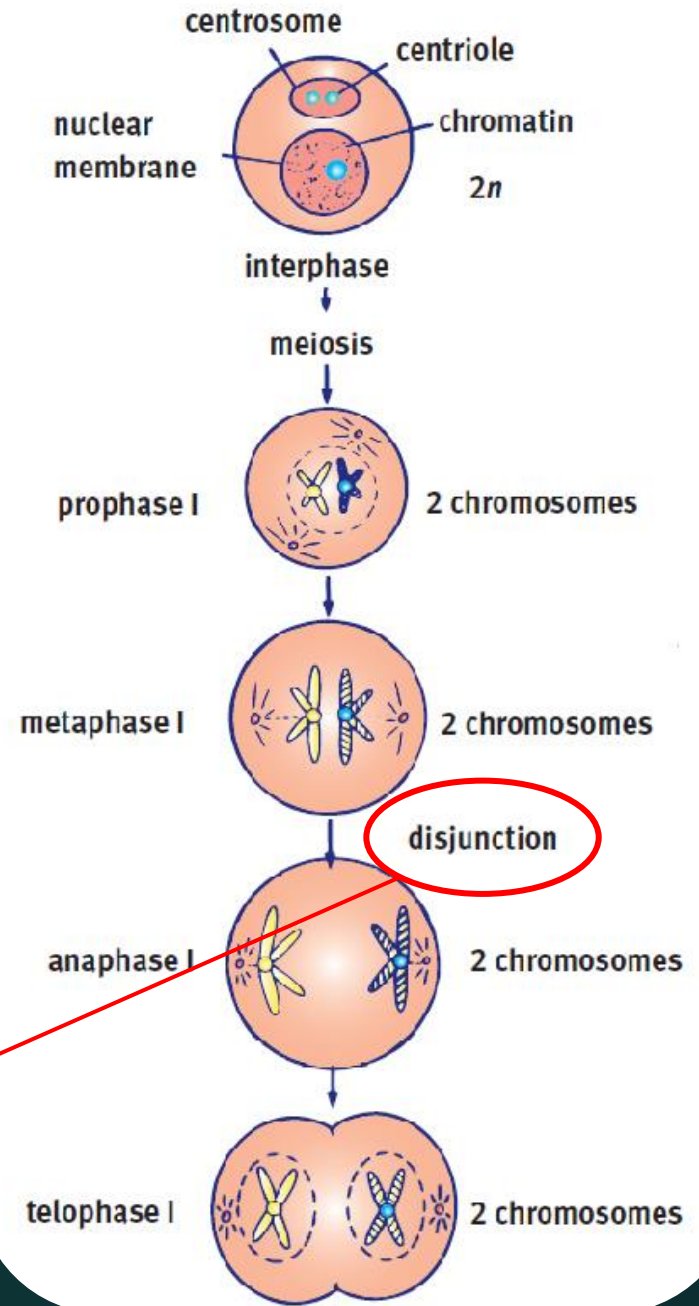
Meiosis

Main differences during stage 1

1. We pull apart homologous chromosomes, not sister chromatids
2. Crossing over occurs



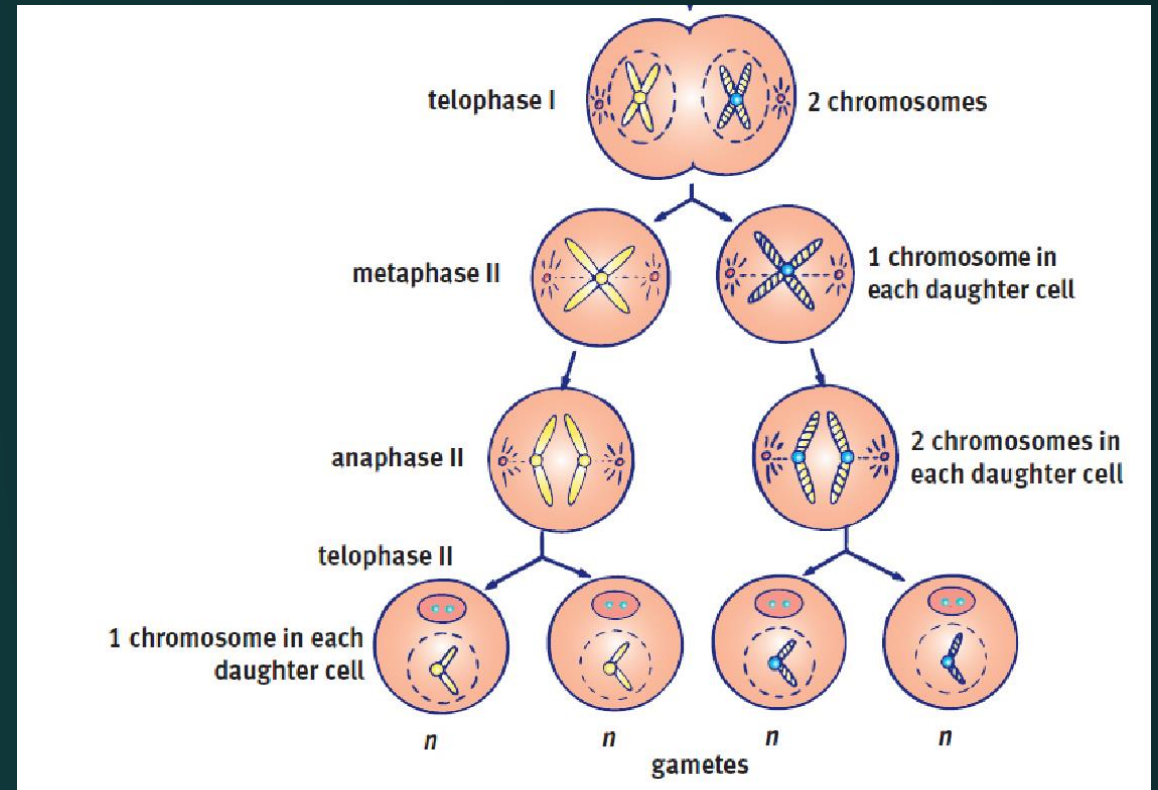
Mendel's First Law of Segregation



Meiosis

Main differences during stage 2

1. NONE!
2. The main difference is we go immediately into stage 2 rather than going around the cell cycle again



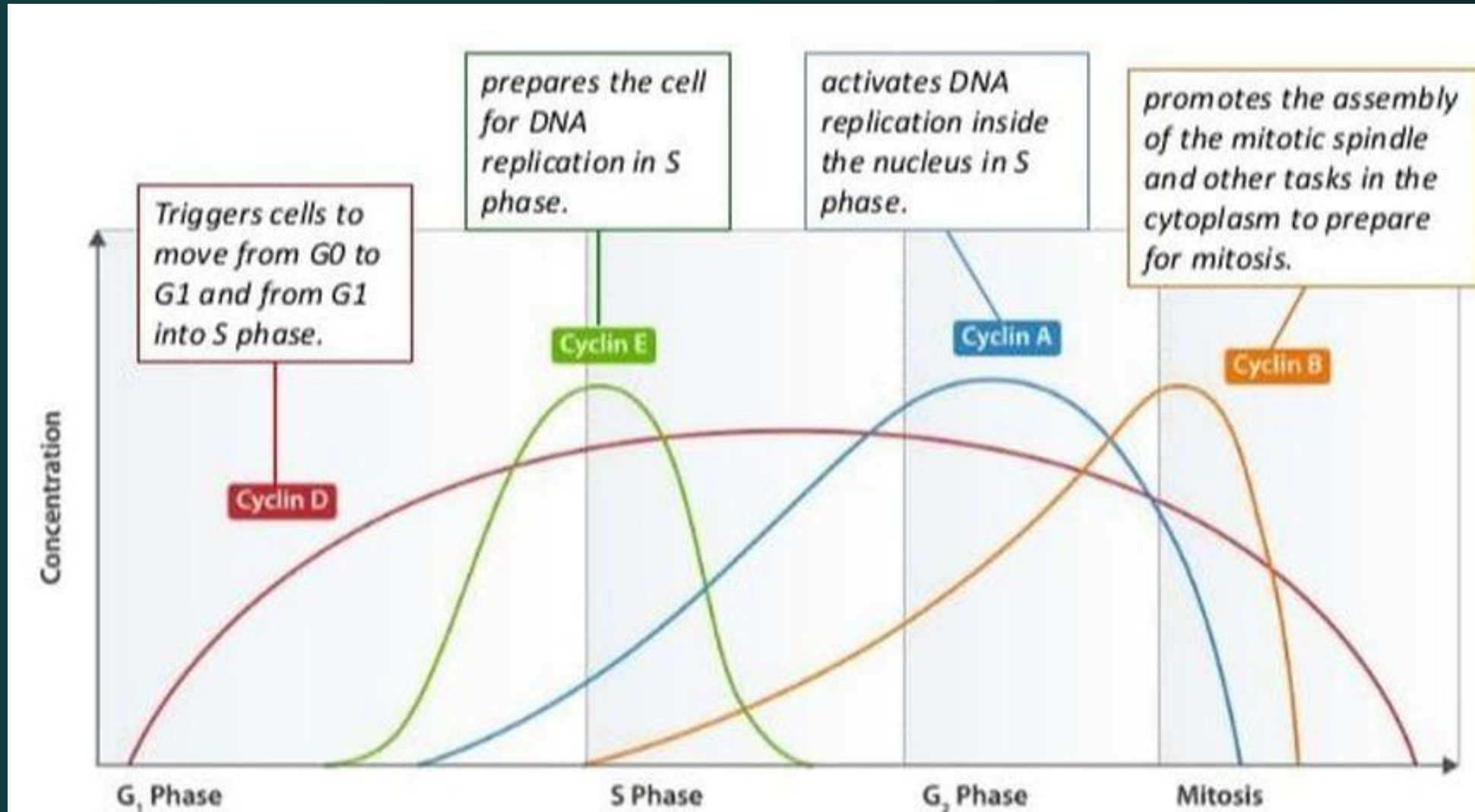
Meiosis

v.

Mitosis

Ploidy	1 $2n \rightarrow 2\ 2n$	1 $2n \rightarrow 4\ 1n$
Cell Specificity	All dividing cells (somatic)	Sex cells (gametes)
Metaphase Pairing	Homologous Chromosomes do NOT pair	Homologous chromosomes pair on opp. sides of metaphase plate
Crossing over?	No	Yes

Aside: Cancer & Cell Cycle Control

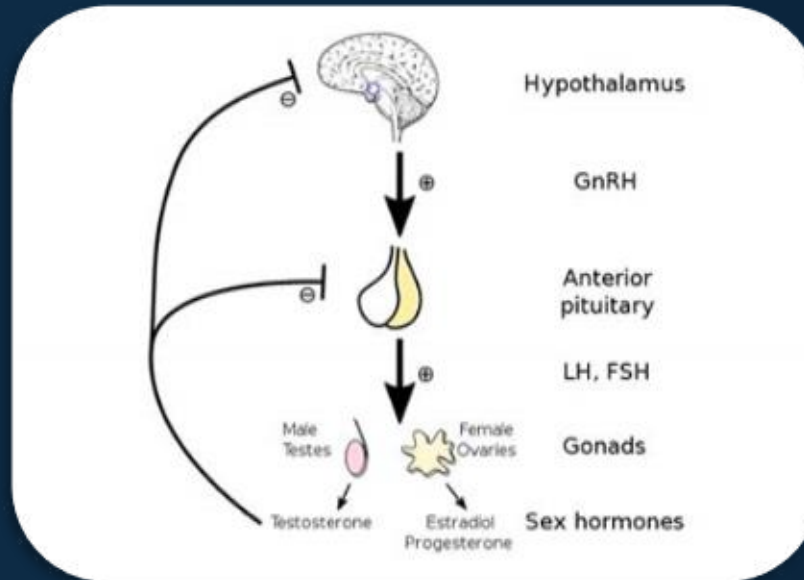


Reproduction

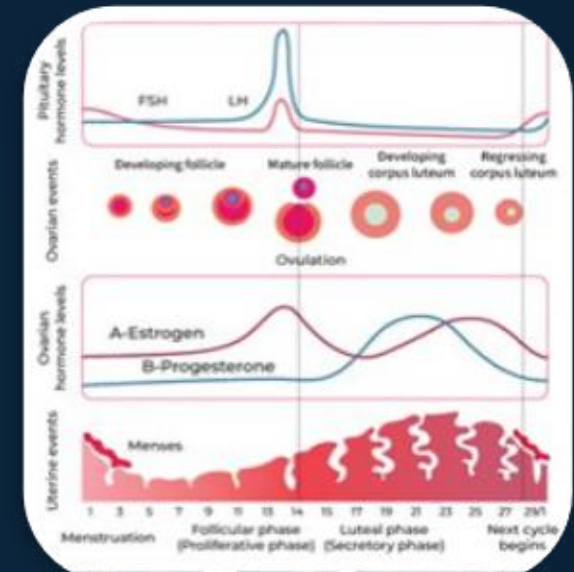
Male and Female Reproductive Systems



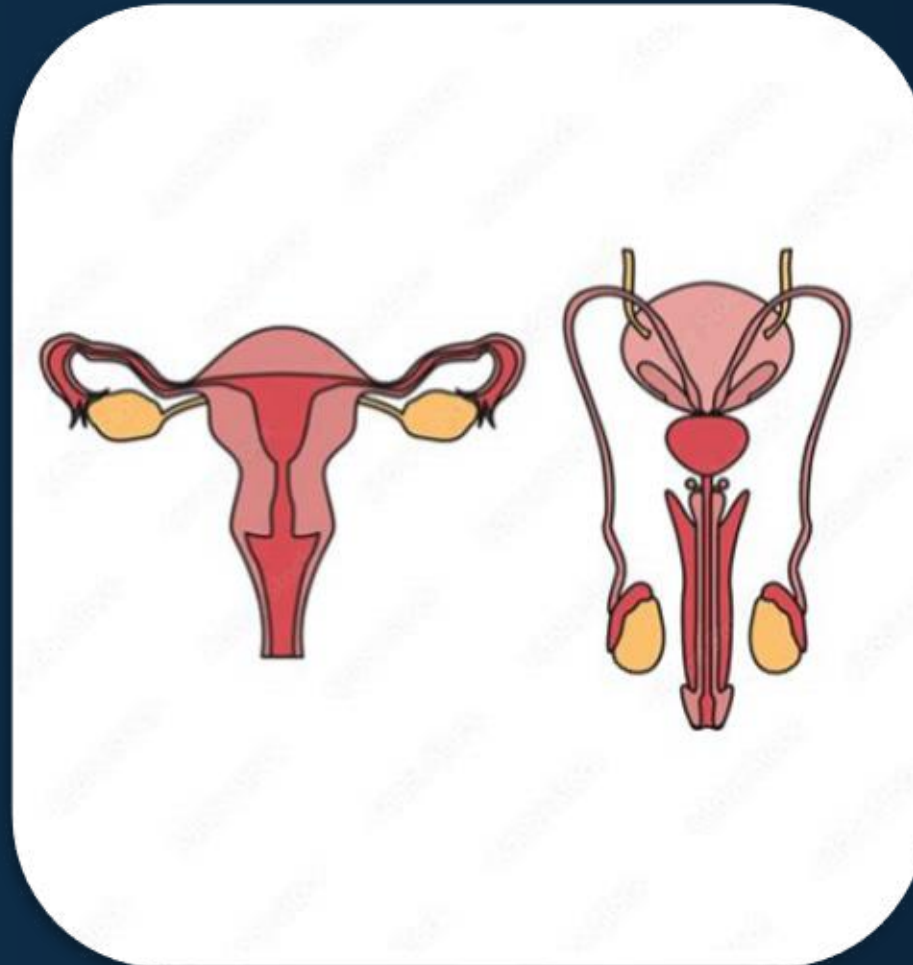
Sexual Development



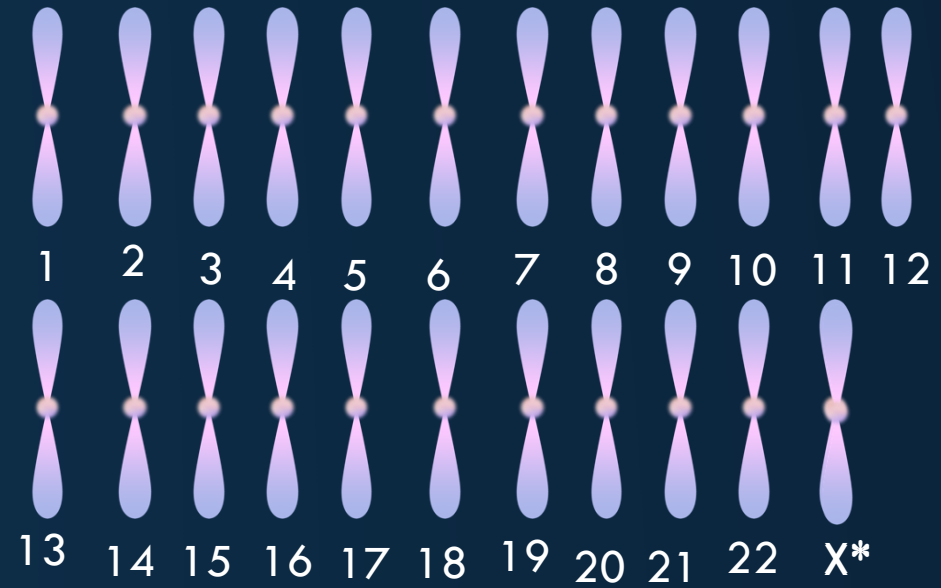
The Menstrual Cycle



Male and Female Reproductive Systems



Chromosomal Sex

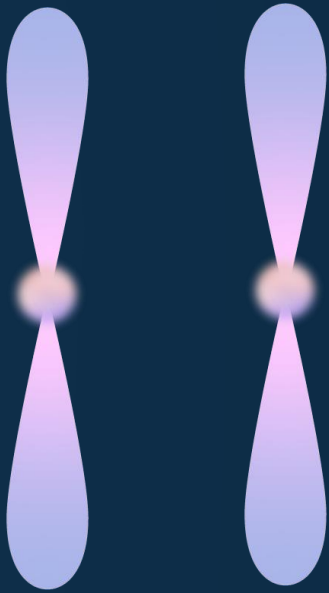


Chromosomal Sex



X*

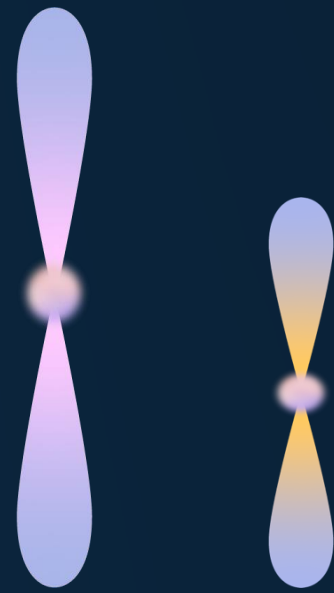
Chromosomal Sex



x

x

Female

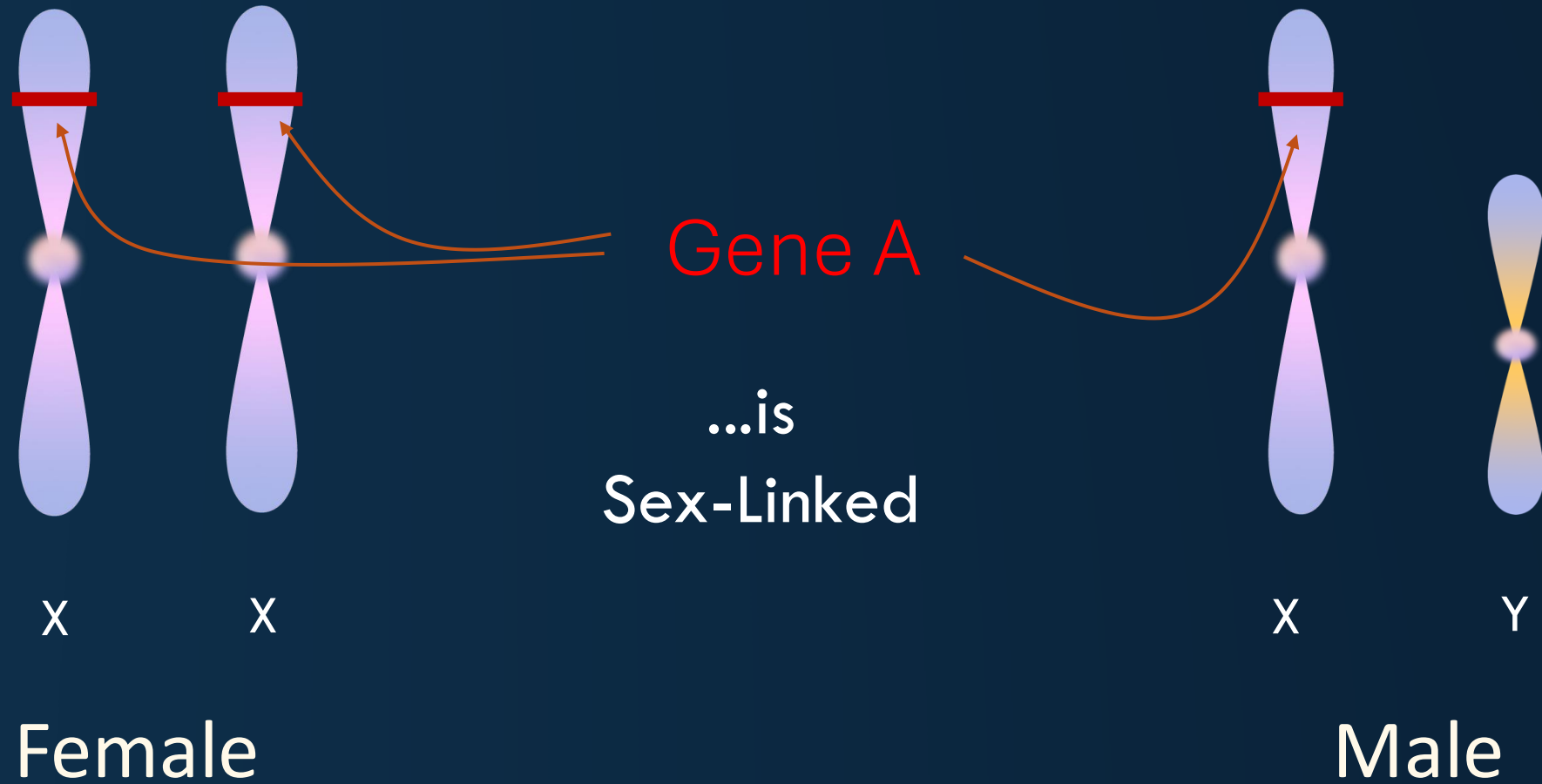


x

y

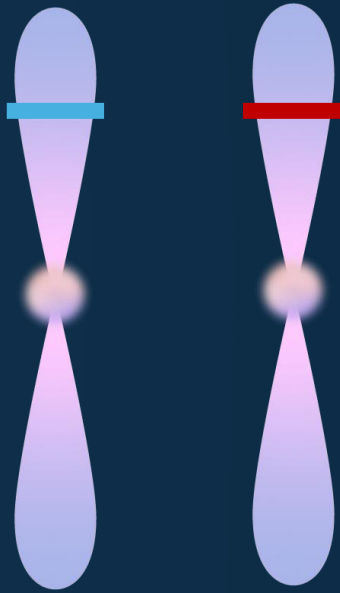
Male

Chromosomal Sex



Chromosomal Sex

Mutant Gene A
(recessive)



X

X

Female



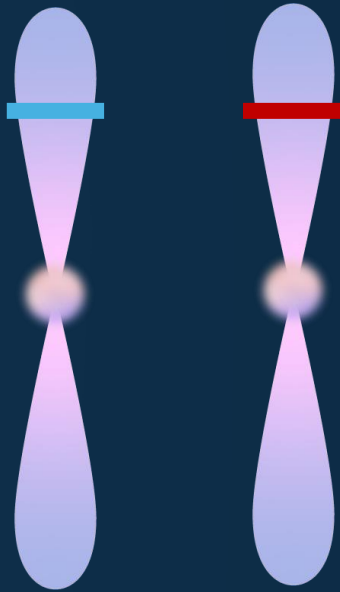
X

Y

Male

Chromosomal Sex

Mutant Gene A
(recessive)



X



X

Carrier,
no disease

Female



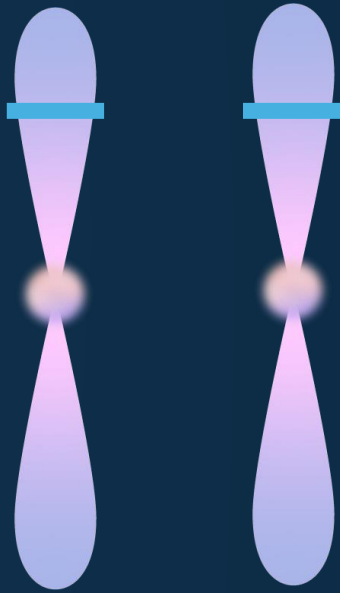
X



Y

Male

Chromosomal Sex



X

X

Female

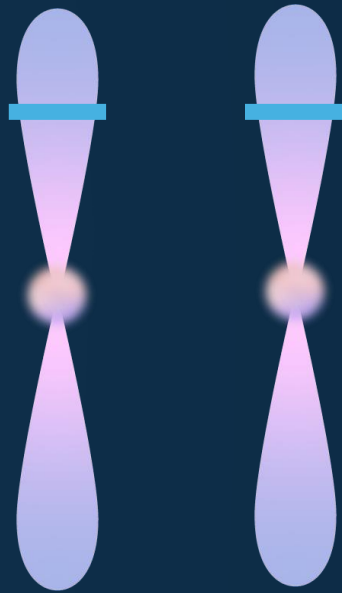


X

Y

Male

Chromosomal Sex



X



X

Female

Has disease



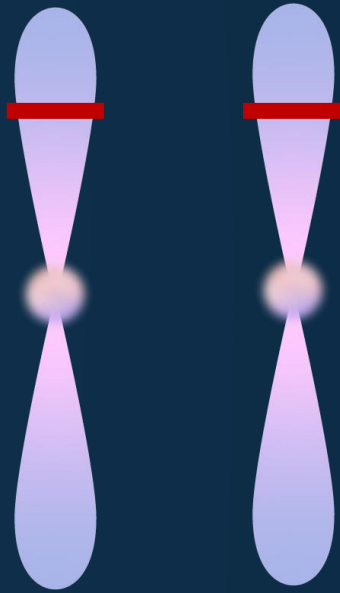
X



Y

Male

Chromosomal Sex



X

X

Female

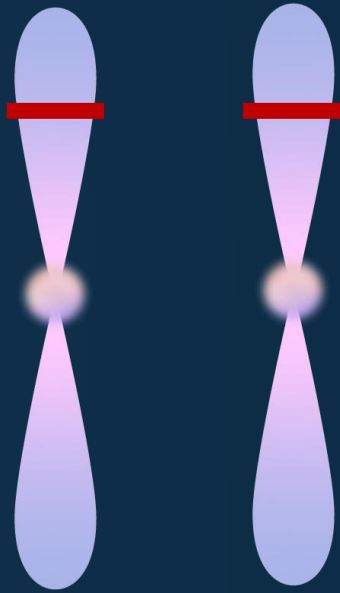


X

Y

Male

Chromosomal Sex



X

X

Female

Has disease



X

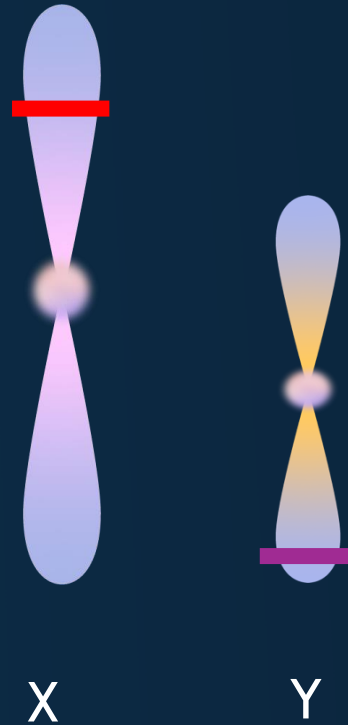
Y

Male

Chromosomal Sex

Most x-linked disorders
are recessive

X-linked disorders are
typically more common in males

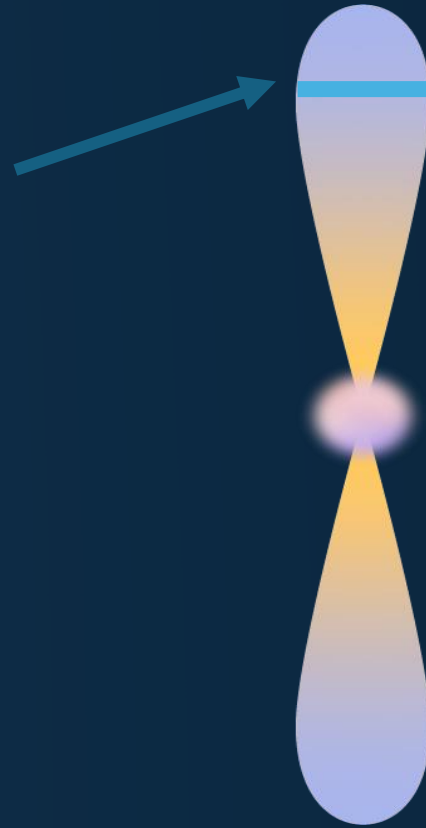


Males are *hemizygous* for genes on the X and Y
chromosomes

Chromosomal Sex

Most notable gene is SRY
(Sex-determining Region Y)

SRY codes for transcription
factor that causes testis (male
gonad) differentiation



Very little genetic information
(only 63 genes)

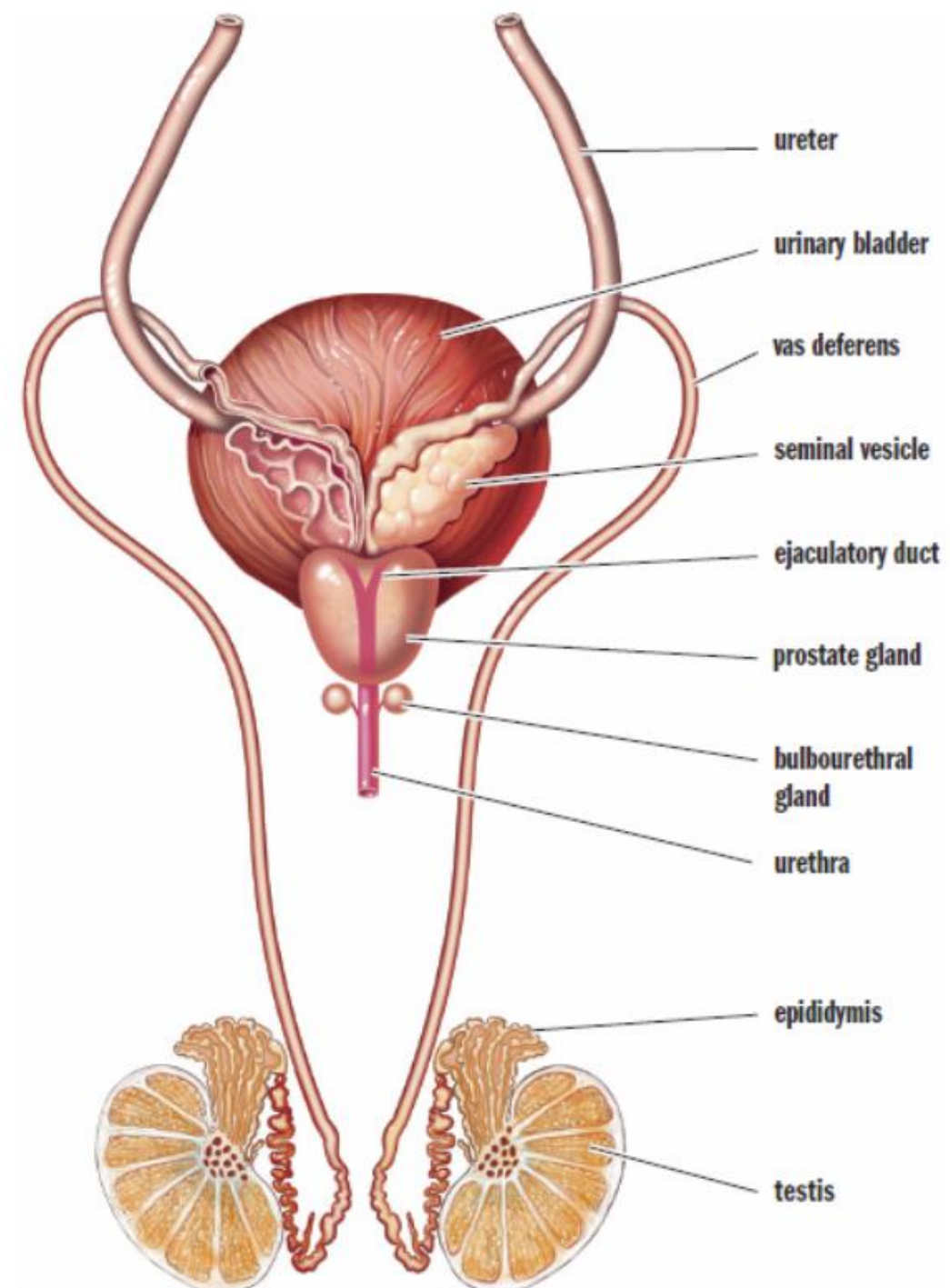
Y

Male Reproductive Anatomy

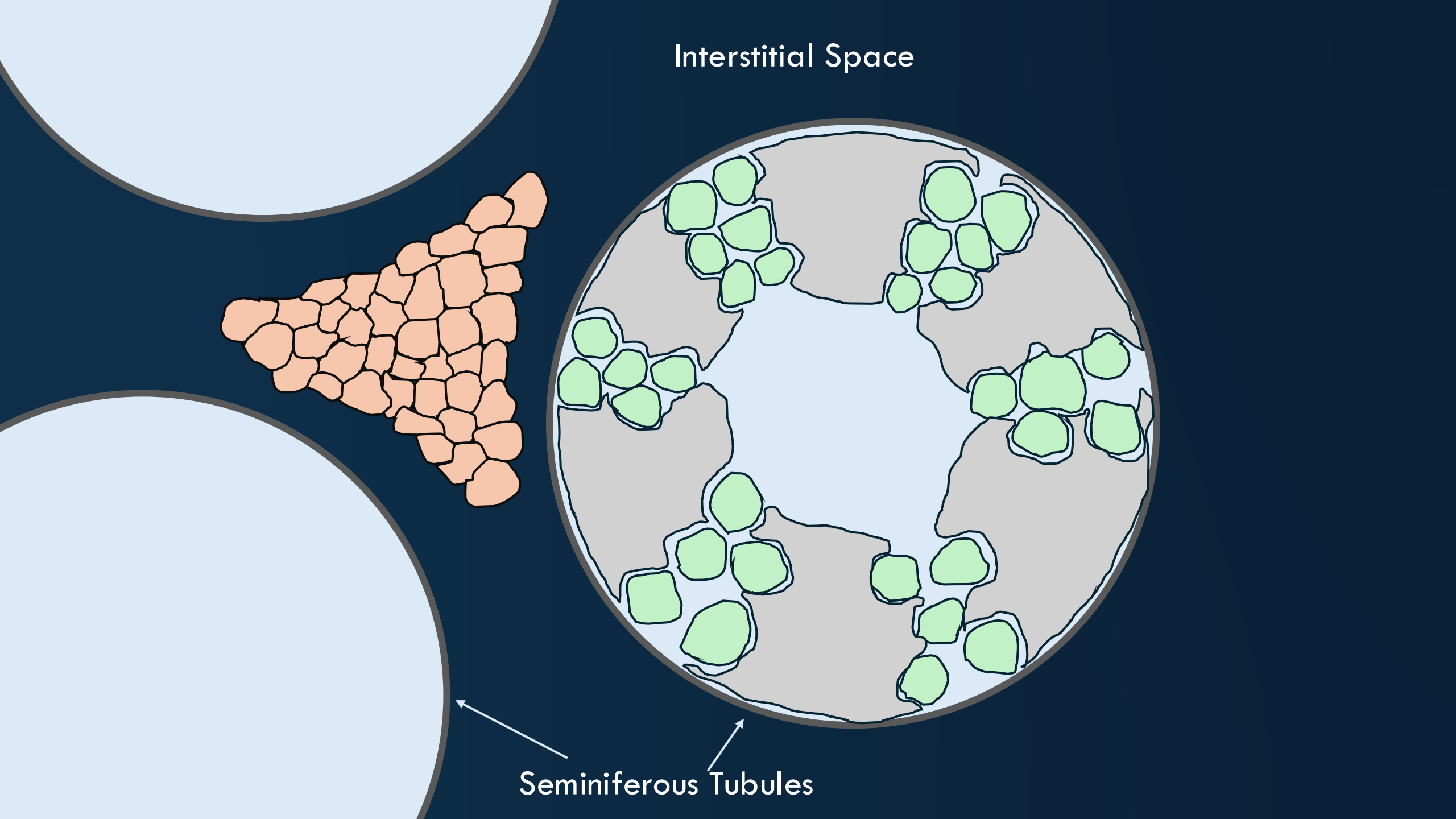
Main purpose: generate and expel sperm

Learning through focusing on function:

we follow the journey of sperm through the reproductive system



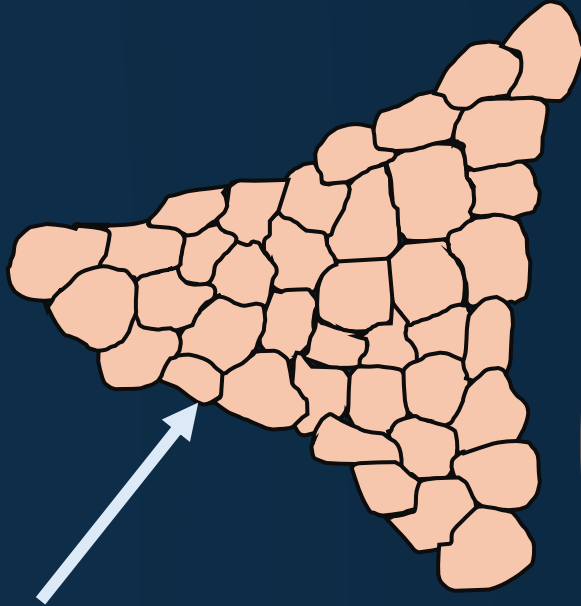




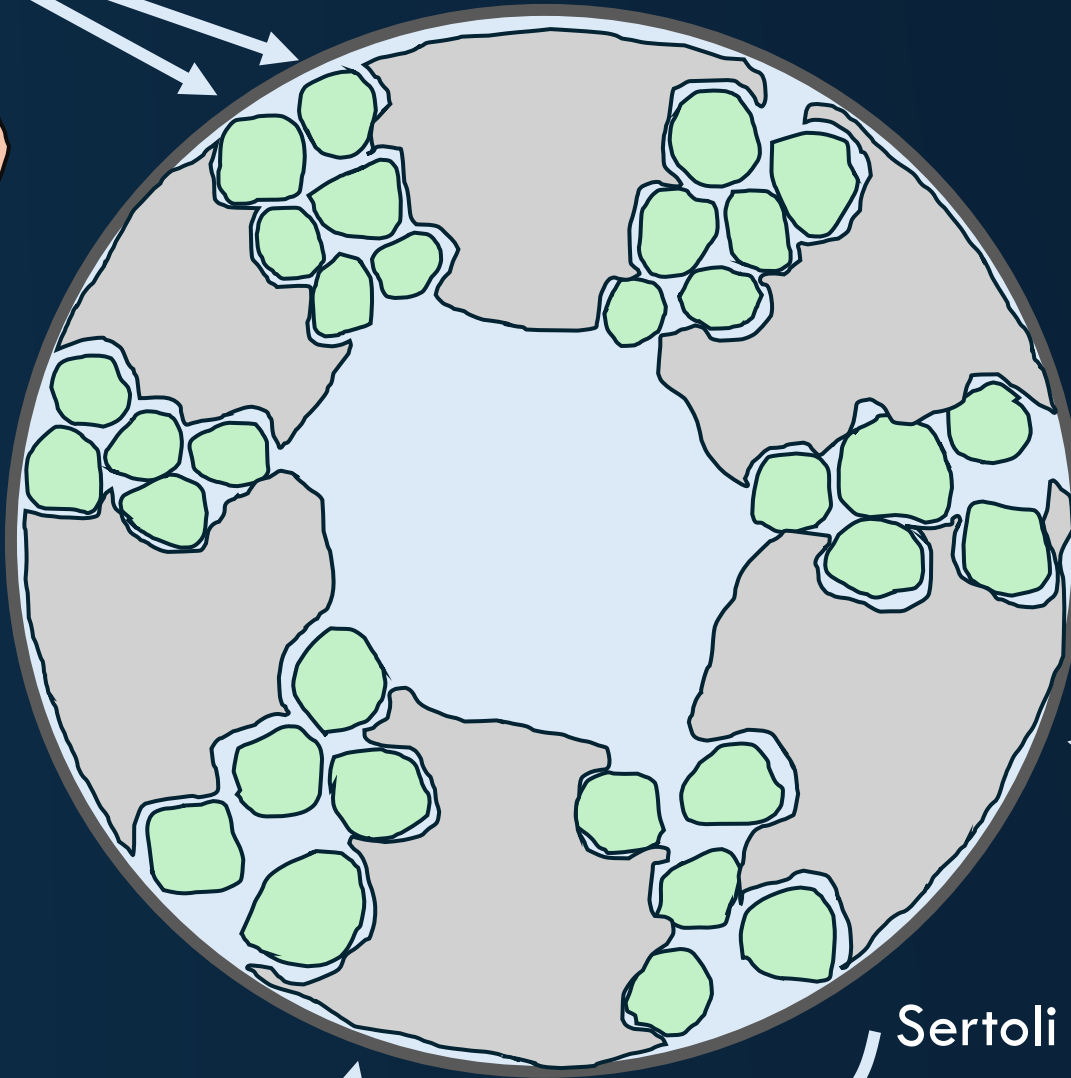
Interstitial Space

Seminiferous Tubules

Germ Cells: Divide and become sperm

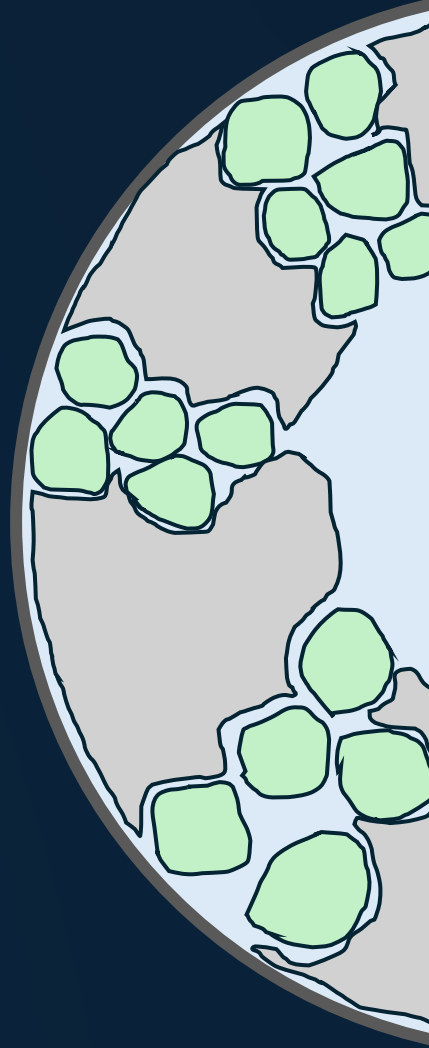
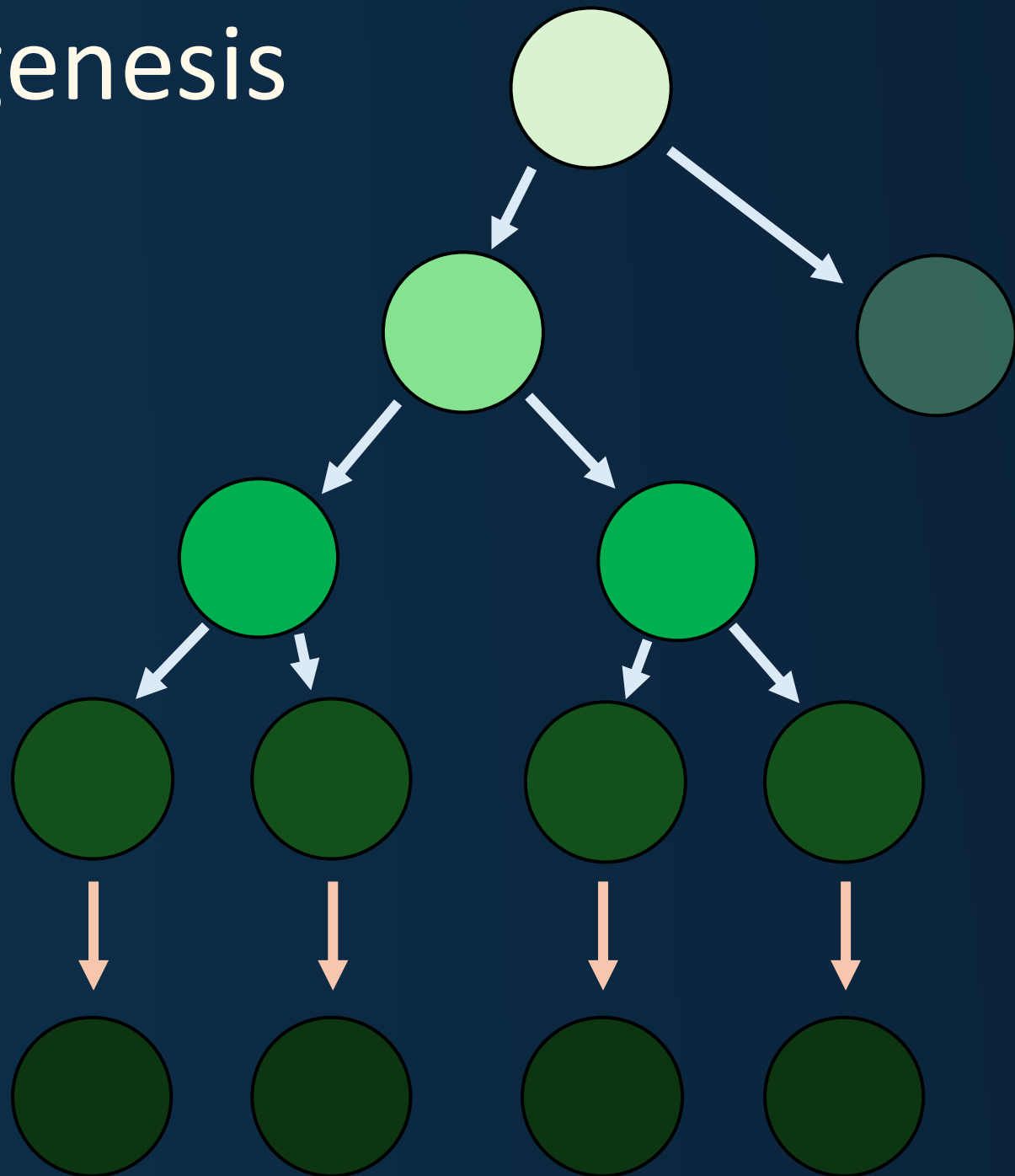


Leydig Cells:
Secrete testosterone and
other androgens



Sertoli Cells:
Nourish developing sperm

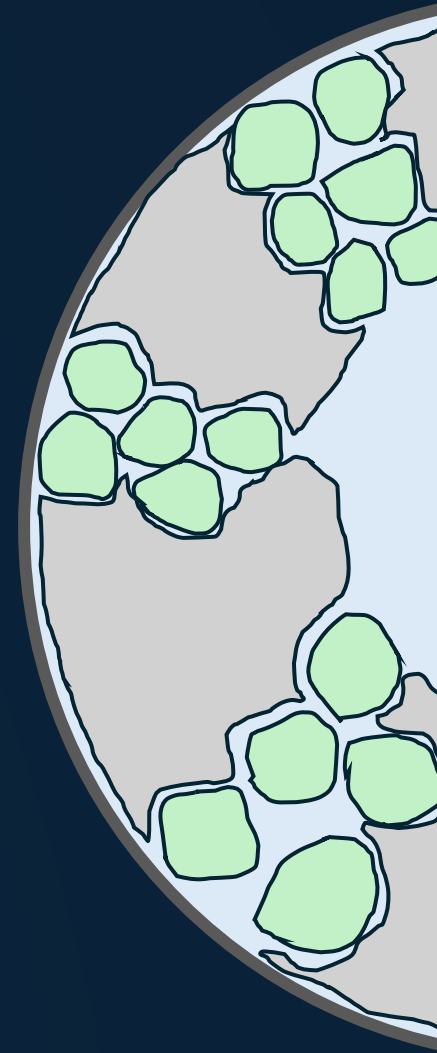
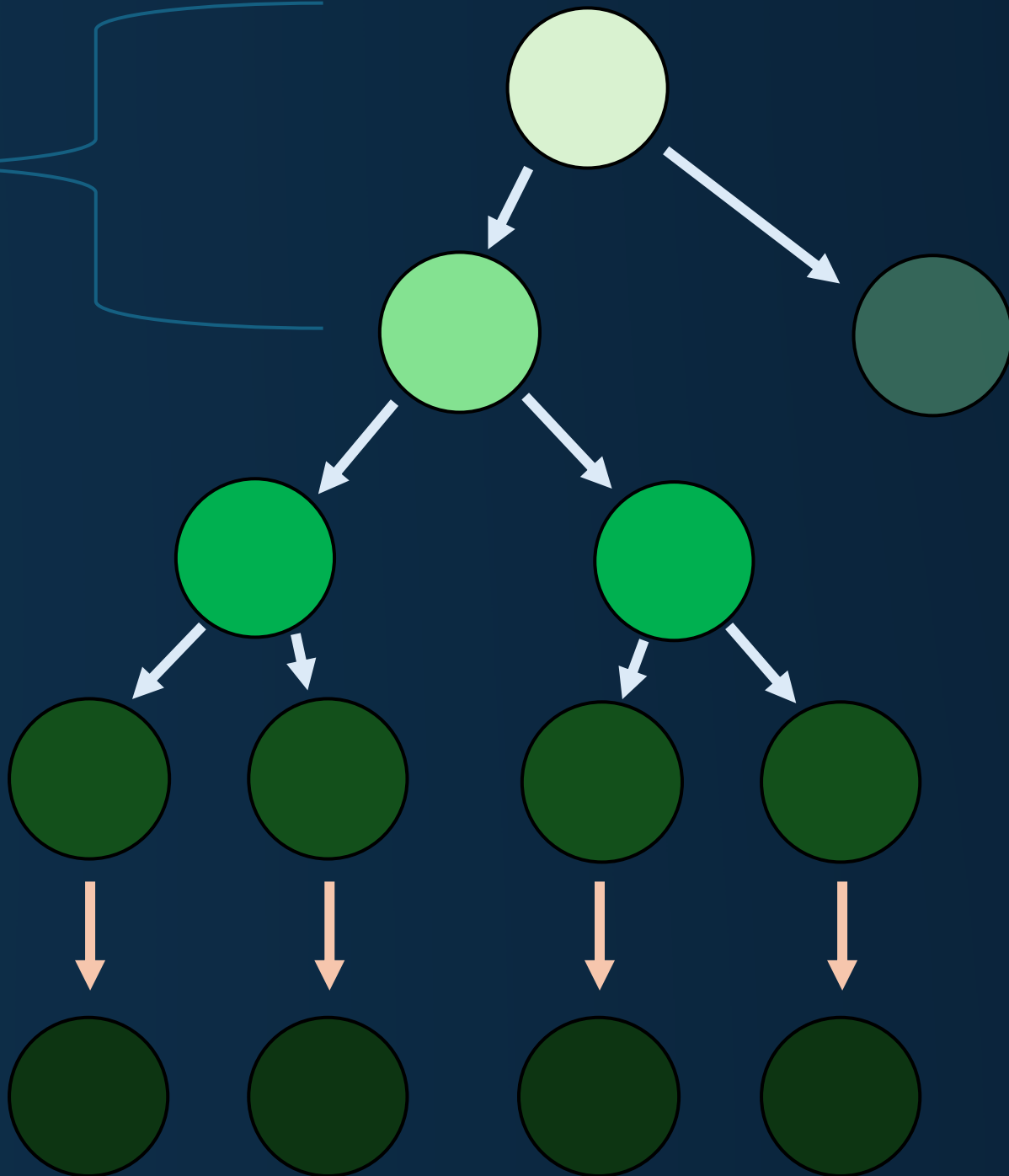
Spermatogenesis



Mitosis

Meiosis

Maturation



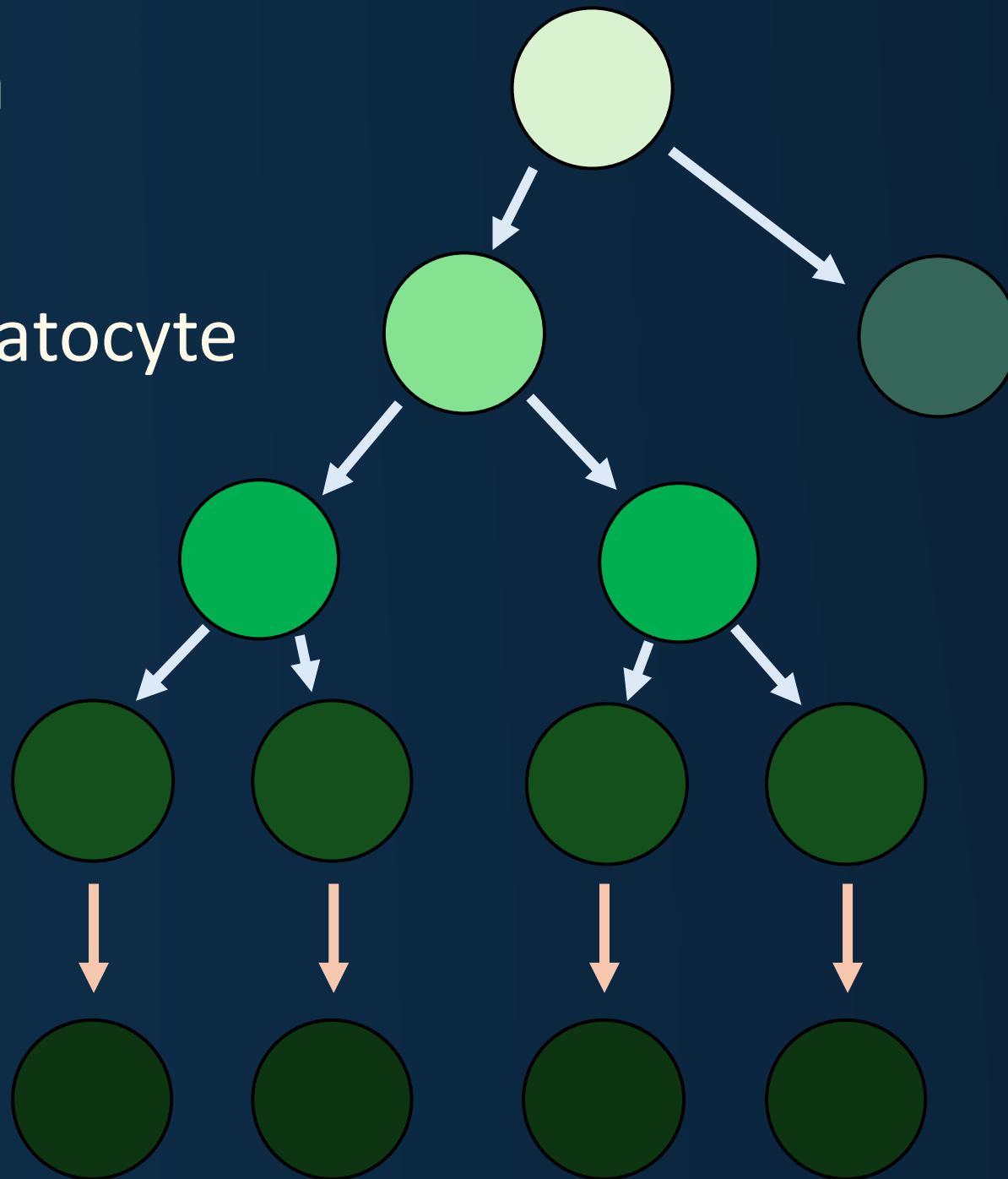
Spermatogonia

Primary Spermatocyte

Secondary
Spermatocyte

Spermatid

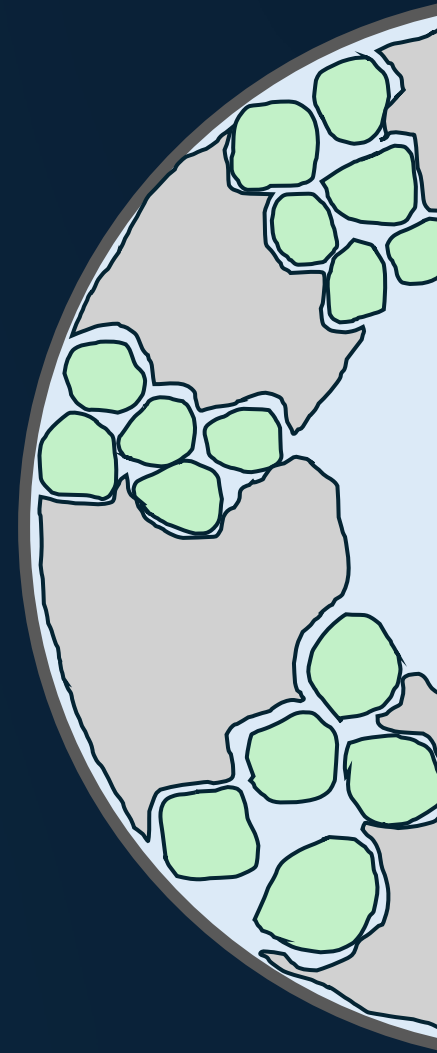
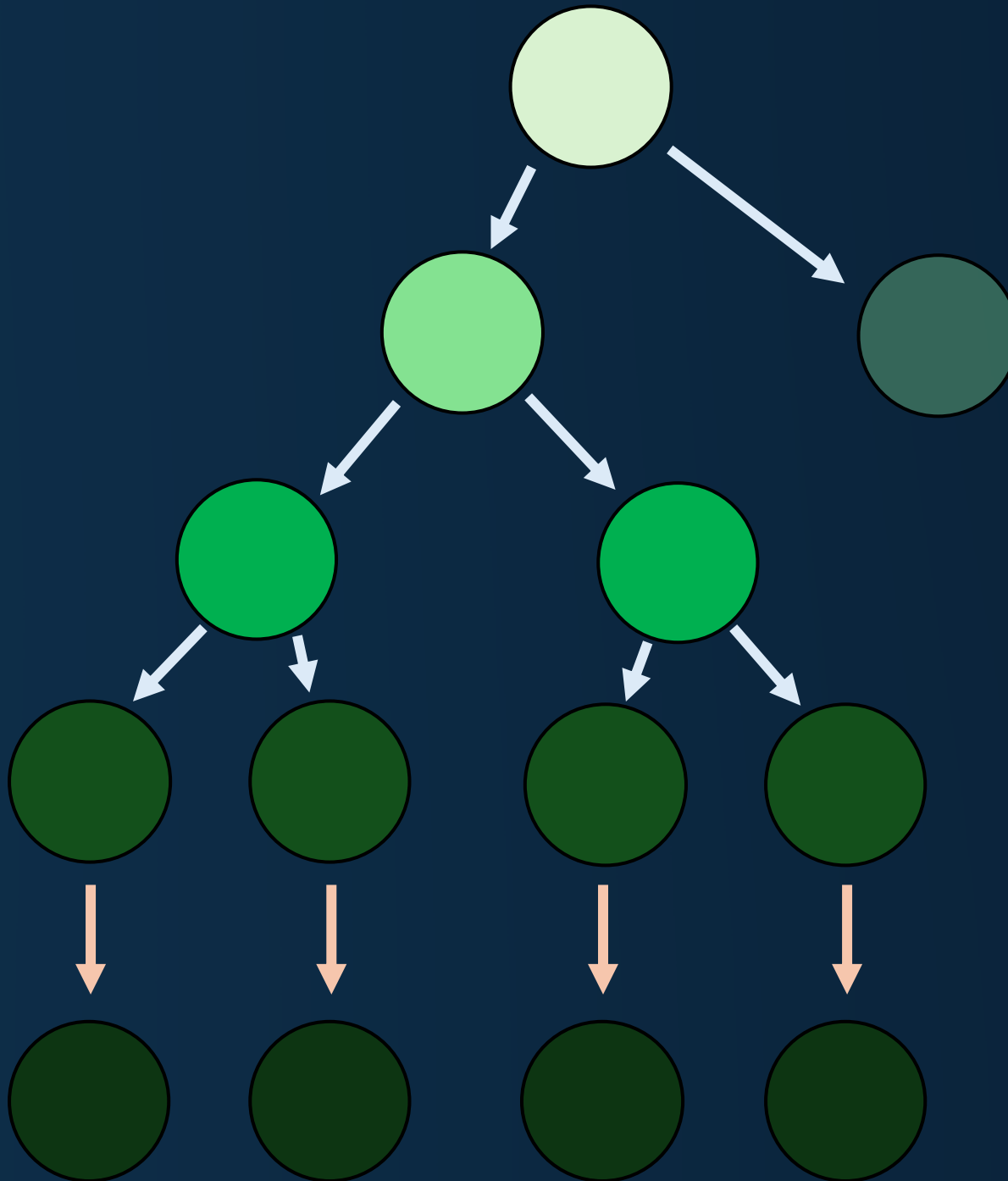
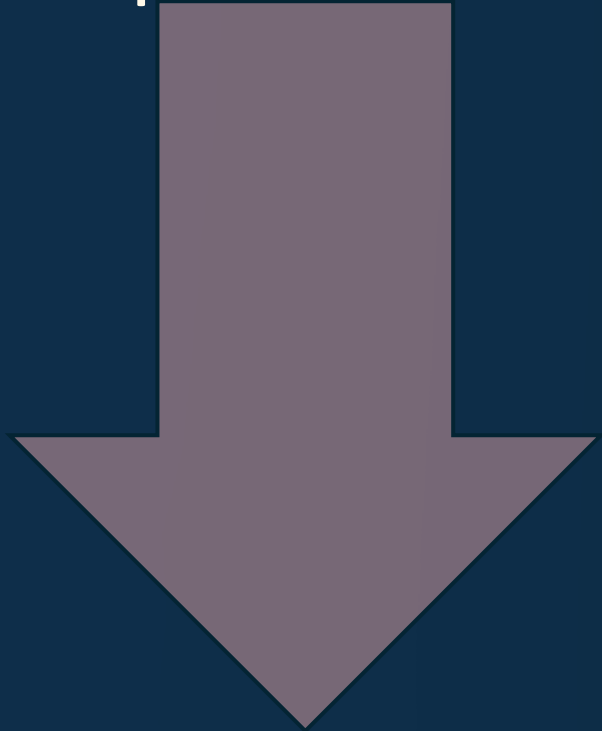
Spermatozoa



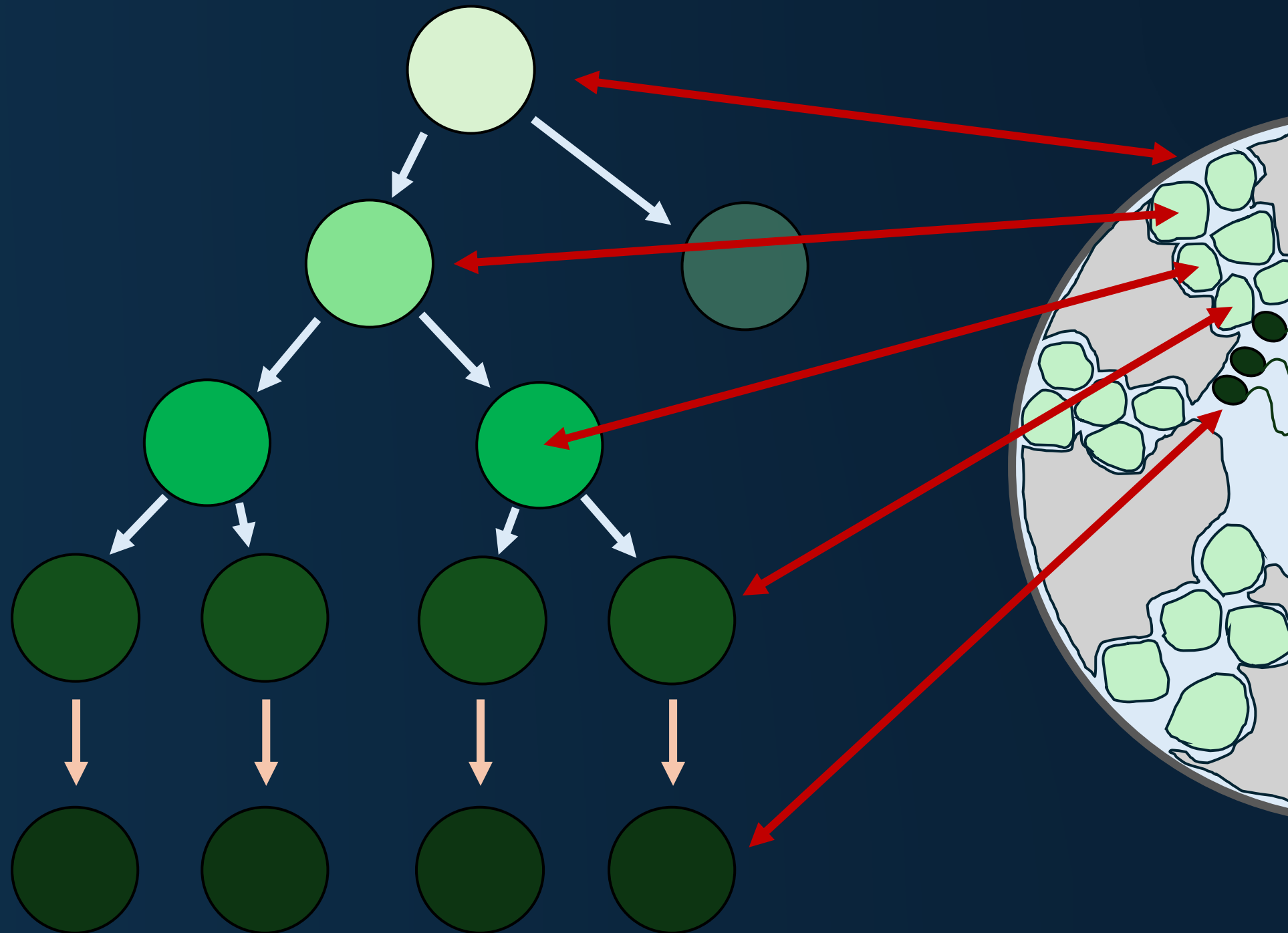
Diploid Cells



Haploid Cells



Edge of Tubule



Lumen

Spermatozoa

Acrosome is important for egg penetration

Midpiece generates energy from fructose

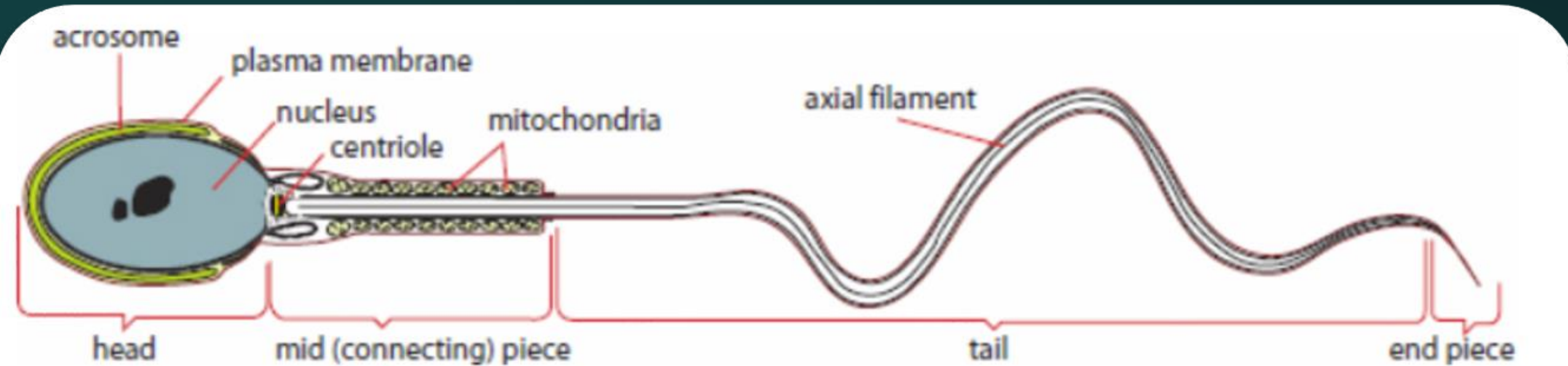


Figure 2.8. Structure of a Mature Sperm

Path of Spermatozoa in Reproductive System

Seminiferous tubules

Epididymis

Vas deferens

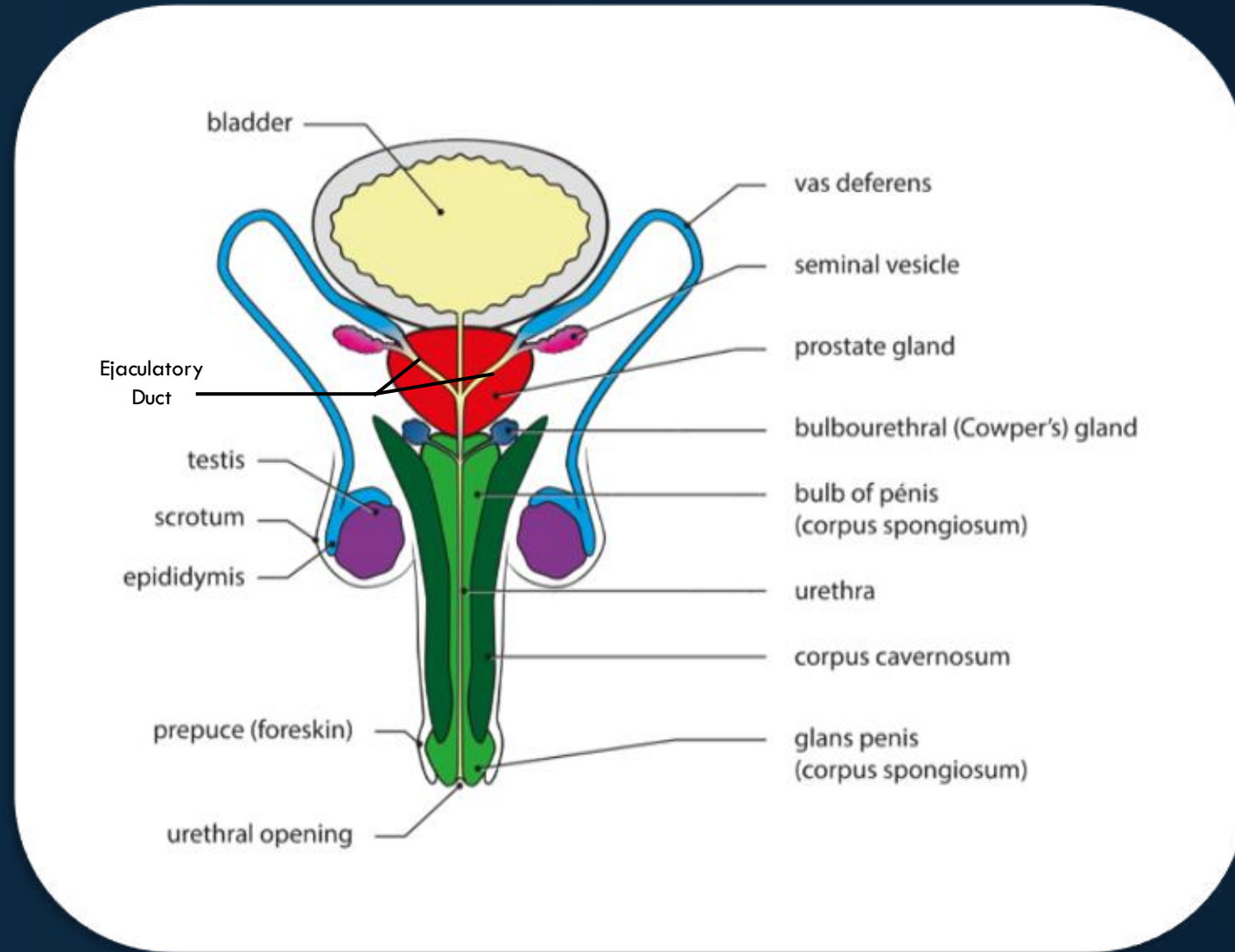
Ejaculatory Duct

(Nothing)

Urethra

Penis

SEVE(N) - UP

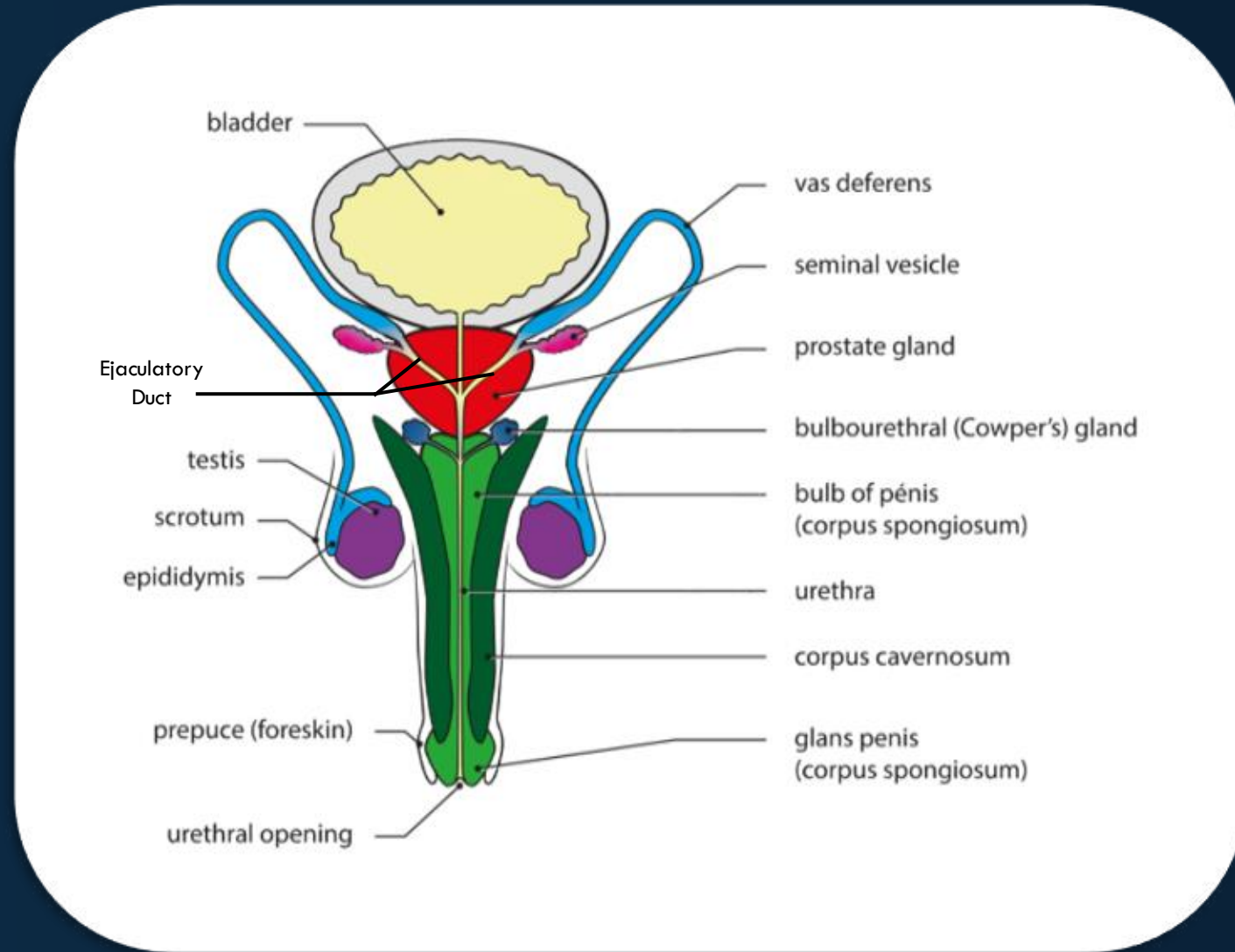


Seminal Fluid-Producing Glands

1. Seminal Vesicle:
Contributes fructose
Makes fluid mildly basic (sperm survival)

2. Prostate Gland:
Also makes fluid mildly basic

3. Bulbourethral/Cowper's Gland:
Secretes viscous fluid, clears remaining
urine, lubricates urethra
(fixes problems of urinary and
reproductive systems sharing urethra)



Path of Spermatozoa, Plus Seminal Fluid Glands

Seminiferous tubules

Epididymis

Vas deferens

Seminal Vesicle

Ejaculatory Duct

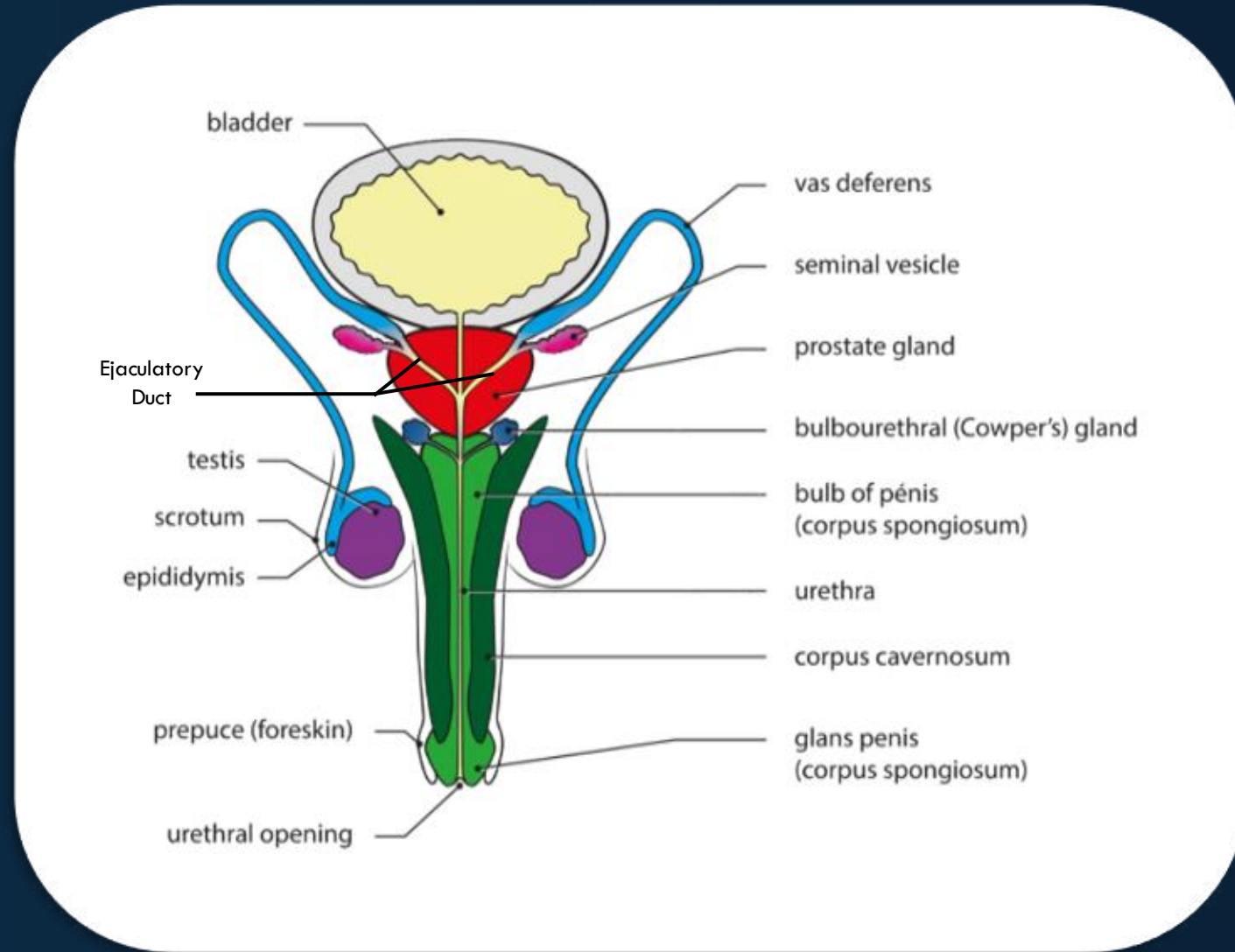
Prostate Gland

(Nothing)

Urethra

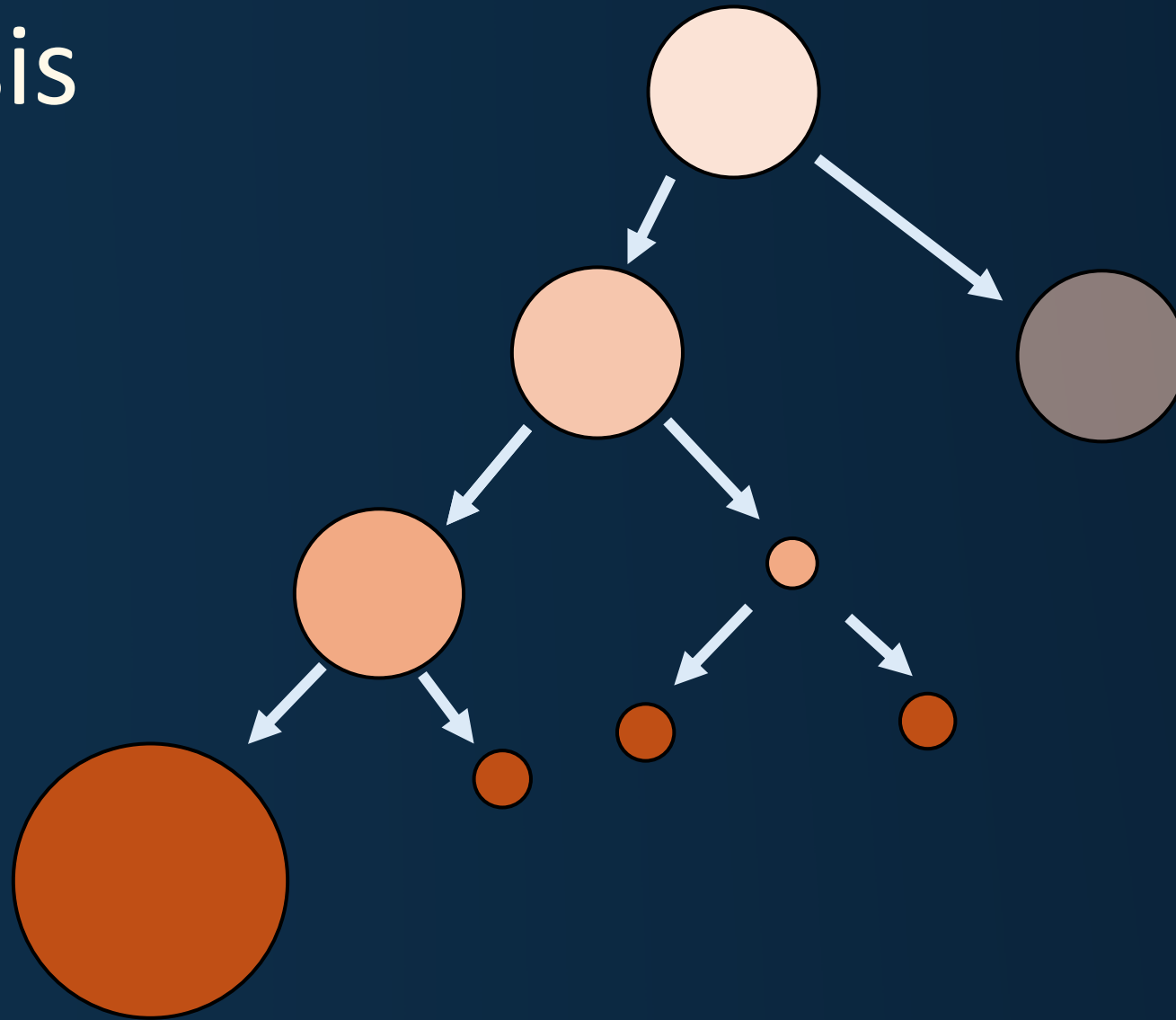
Bulbourethral/Cowper's Gland

Penis



Seminal Fluid + Sperm = Semen

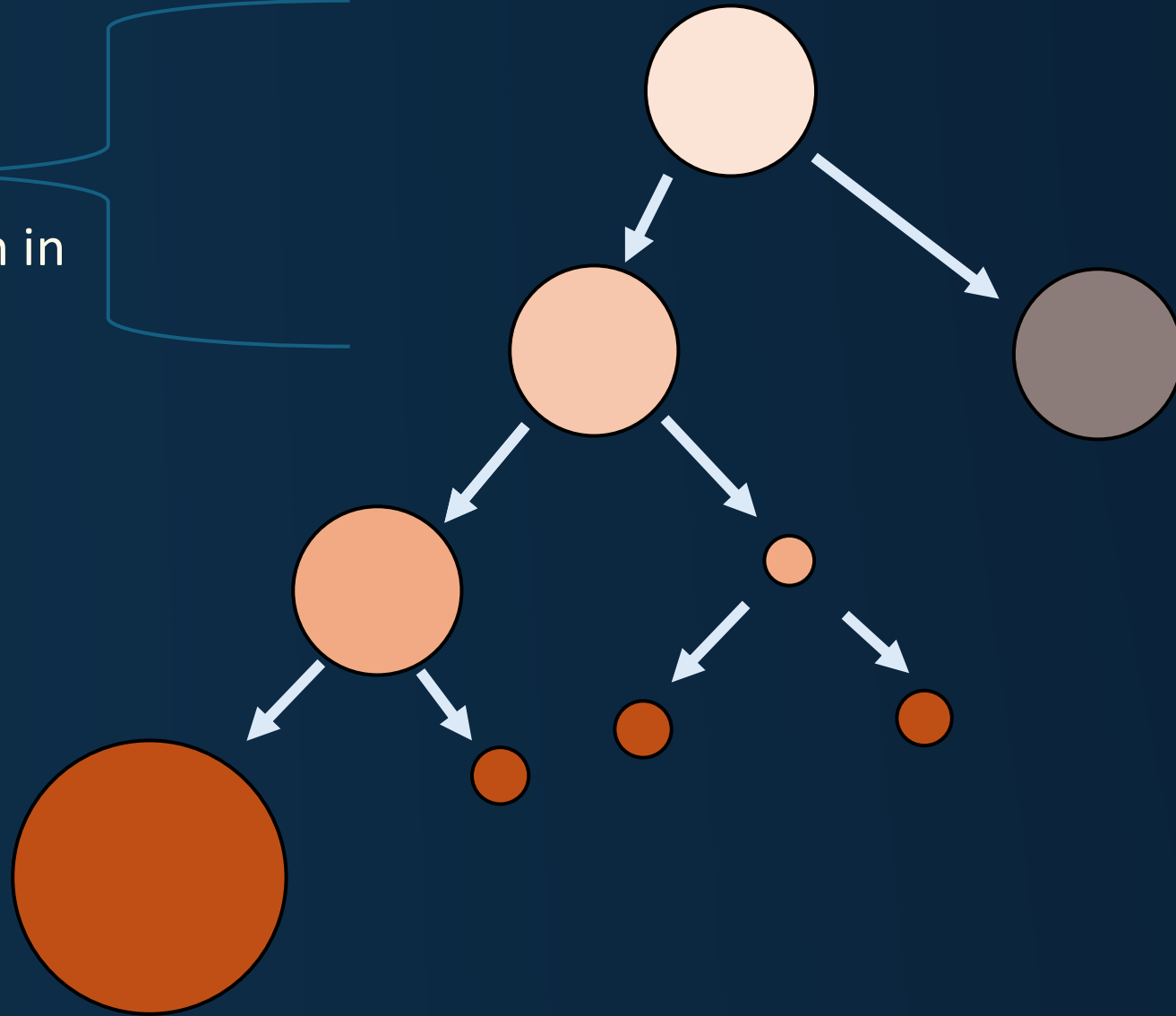
Oogenesis



Mitosis

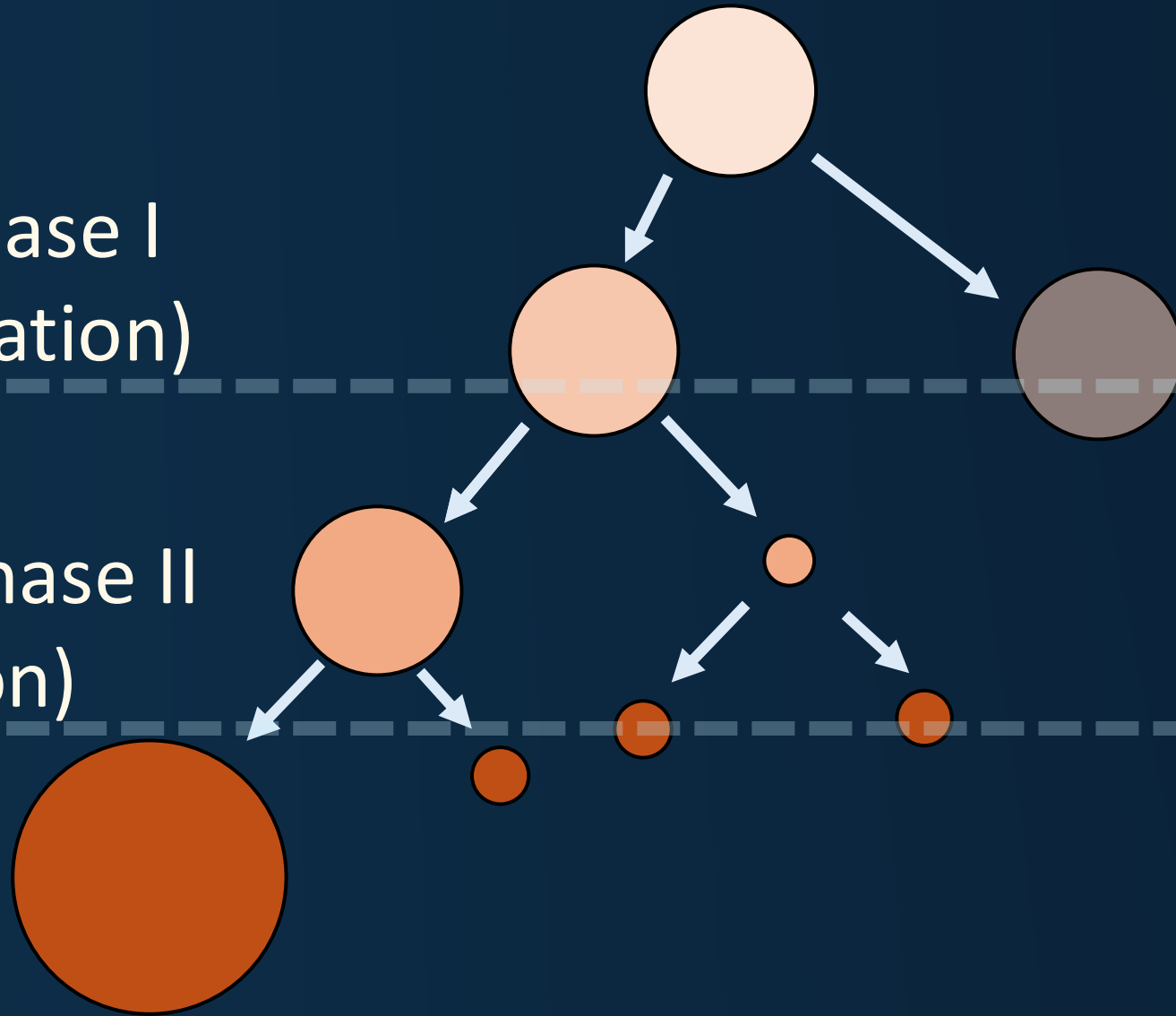
(Completed Before Birth in ovaries)

Meiosis



Arrest in Prophase I
(Until Menstruation)

Arrest in Metaphase II
(Until Fertilization)

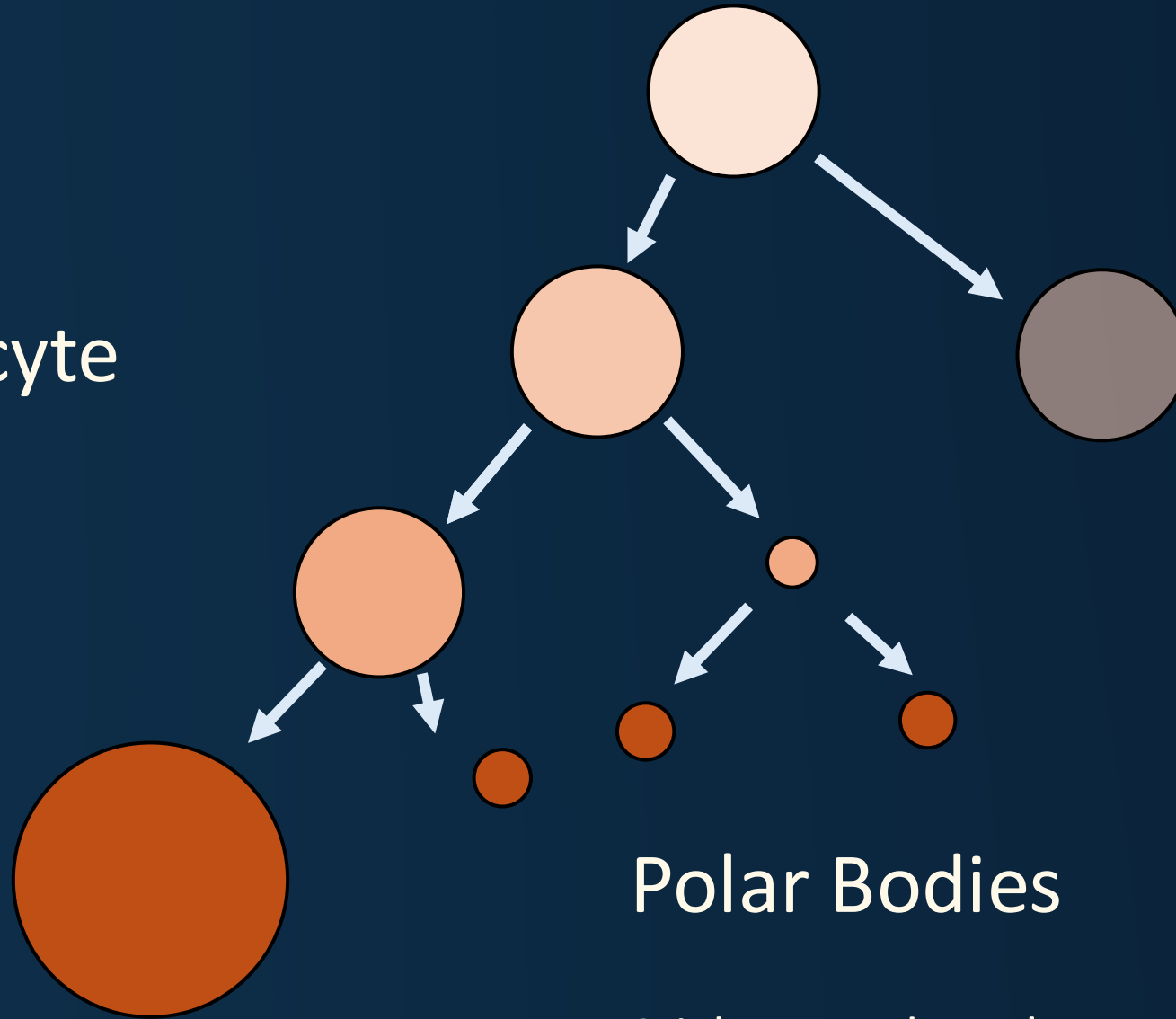


Oogonia

Primary Oocyte

Secondary
Oocyte

Mature
Ovum



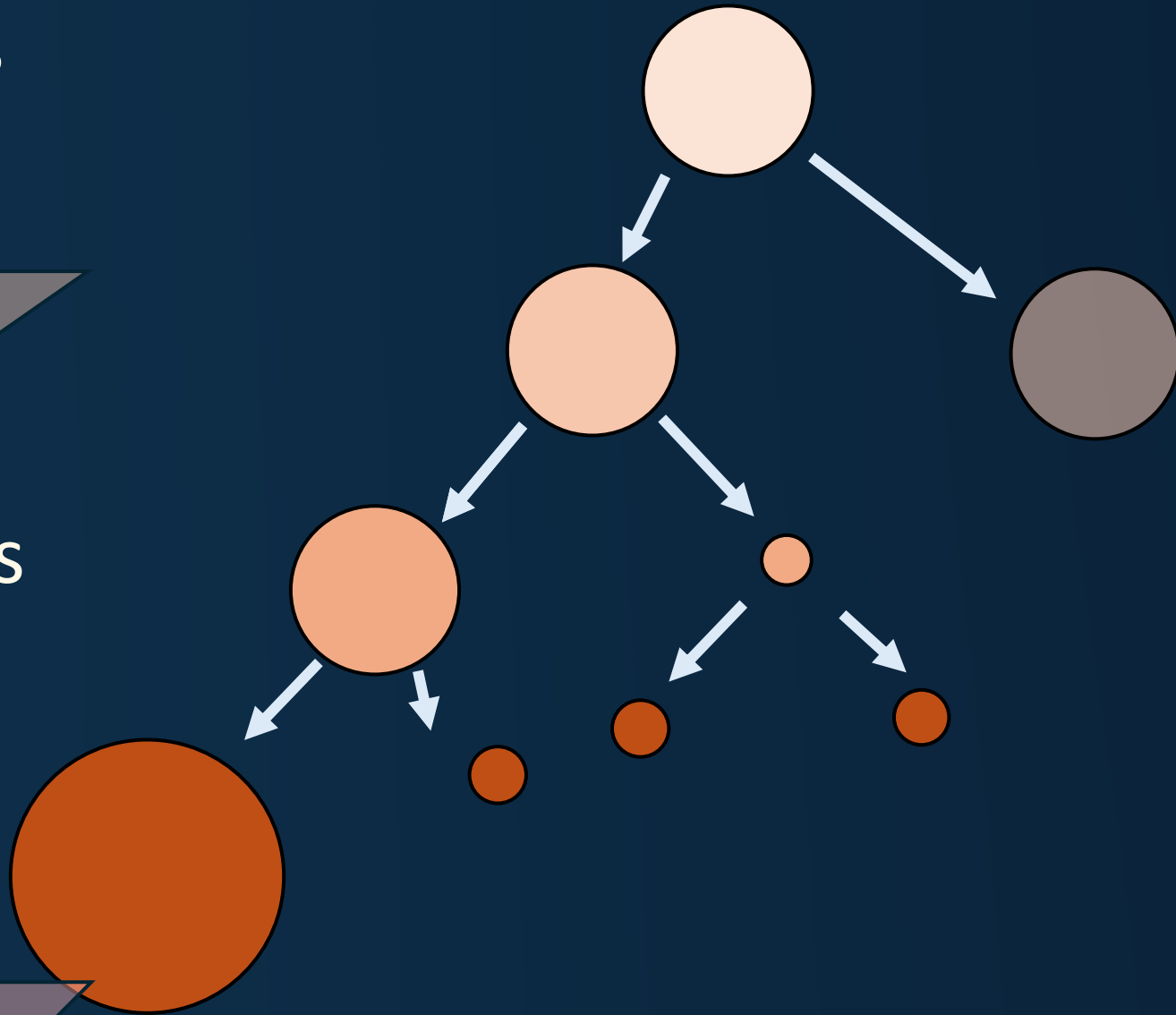
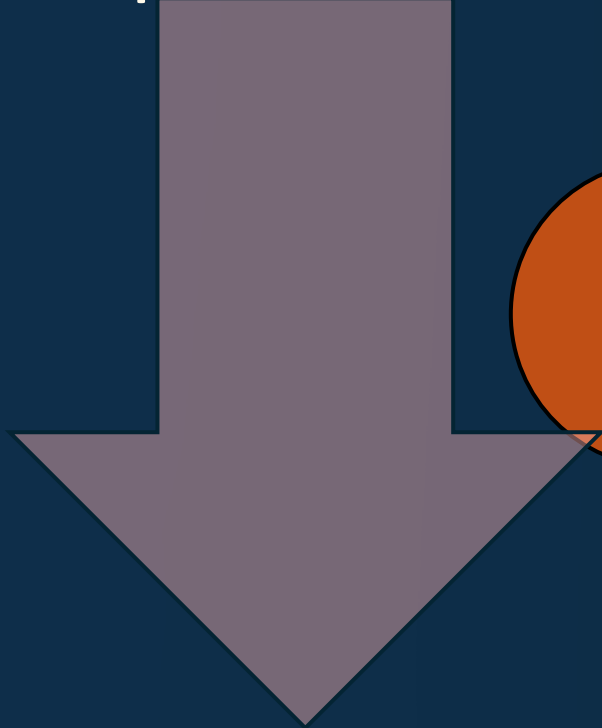
Polar Bodies

Stick around on the egg until disintegration

Diploid Cells



Haploid Cells



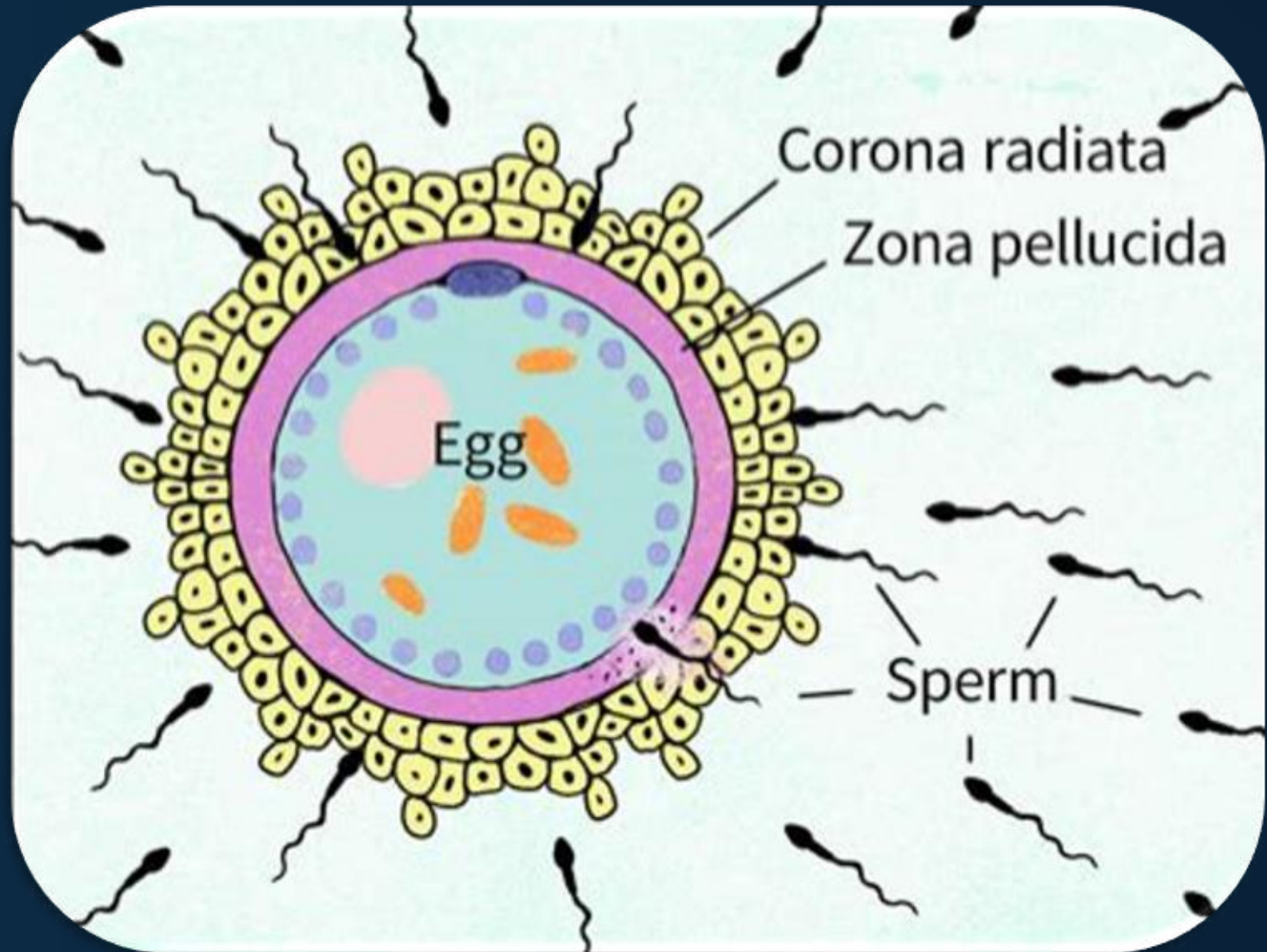
Secondary Oocyte

Zona Pellucida – mix of glycoproteins, necessary for sperm cell binding

Corona Radiata– Layer of cells that stick to oocyte

Sperm must penetrate both of these layers

Upon sperm penetration, oocyte will finally divide into mature ovum.



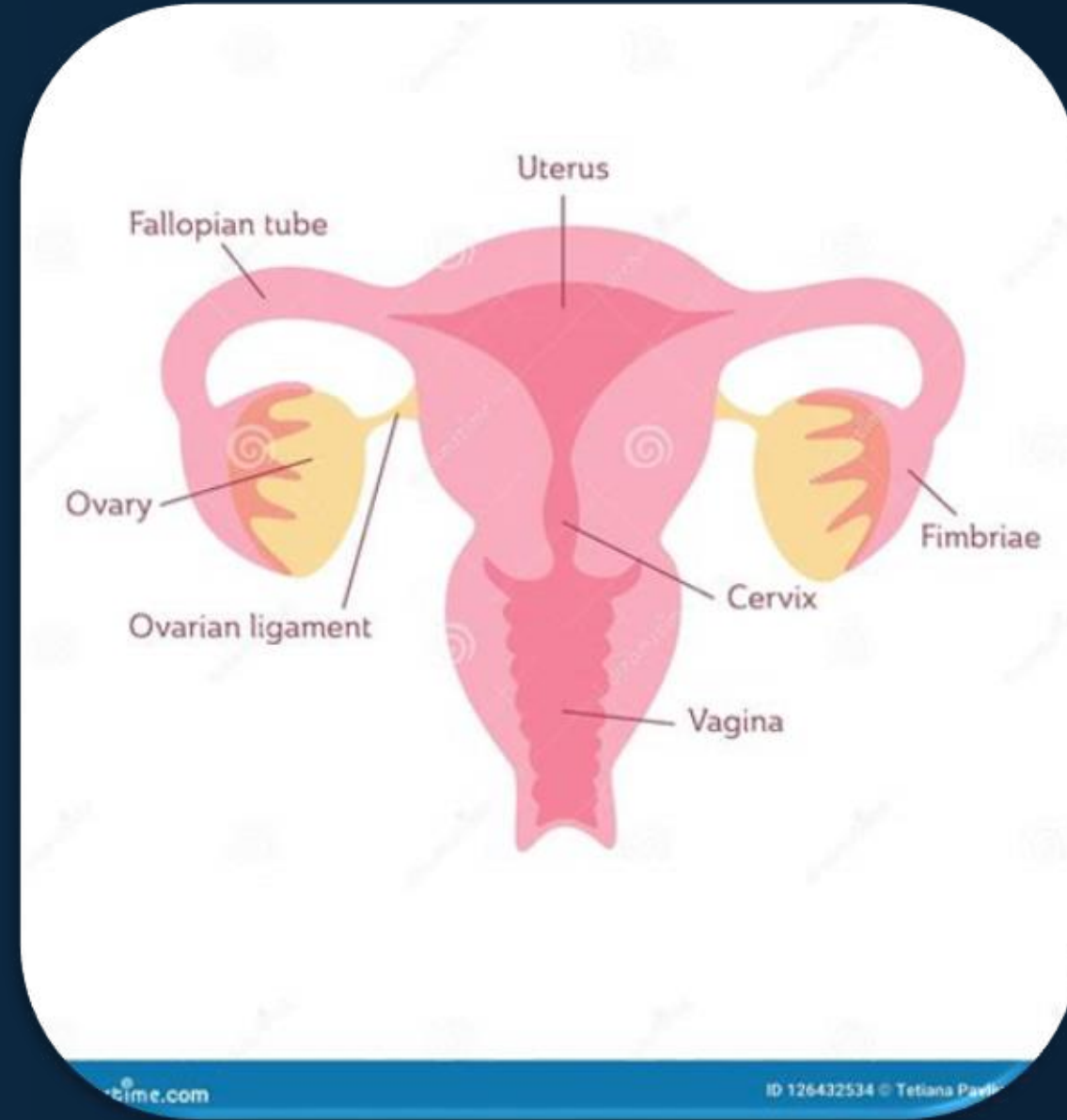
Path of Oocyte in Reproductive System

Every month, only one Primary Oocyte is released from arrest (meiosis I occurs)

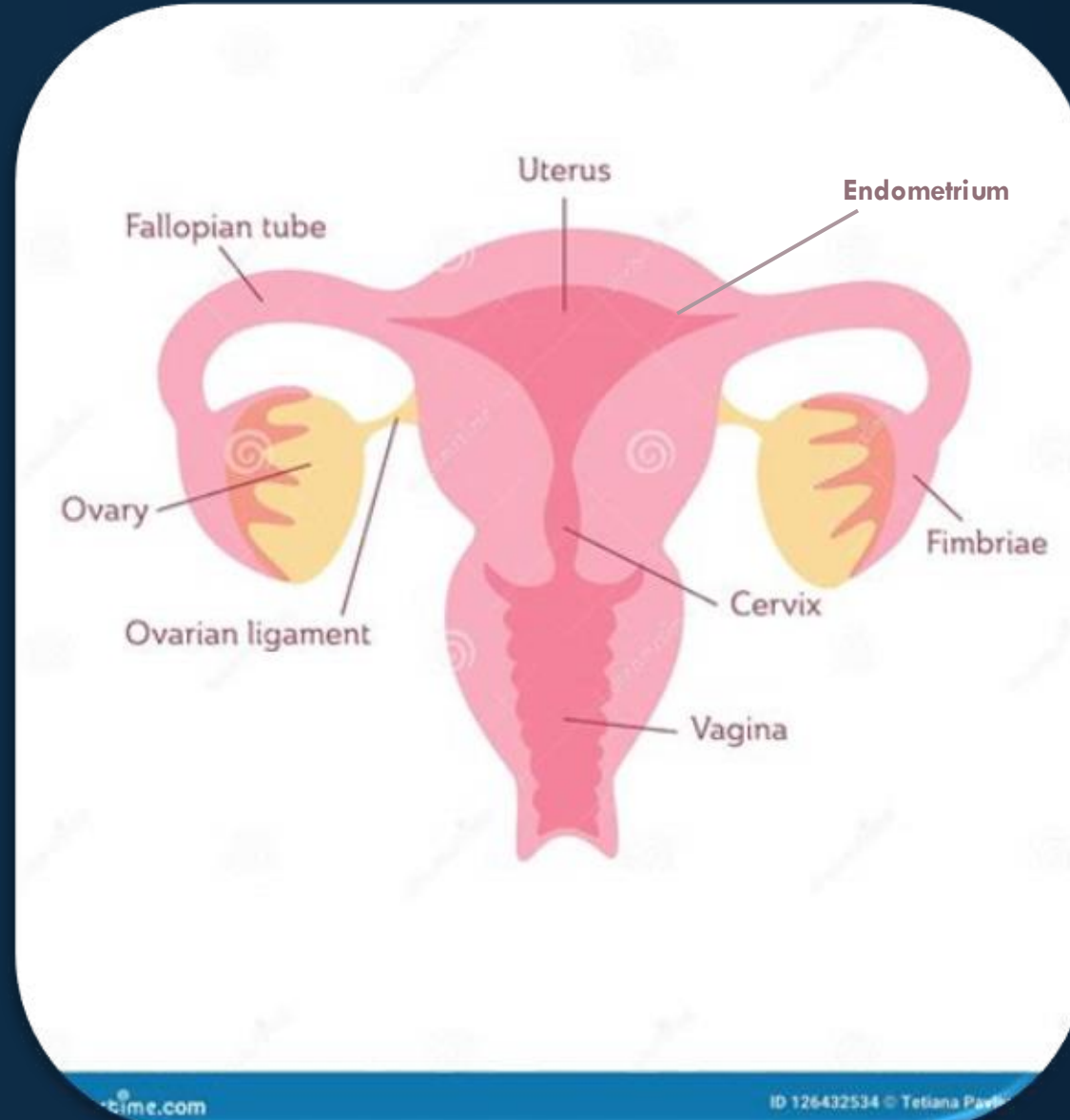
Oocyte is released from ovary into peritoneal space

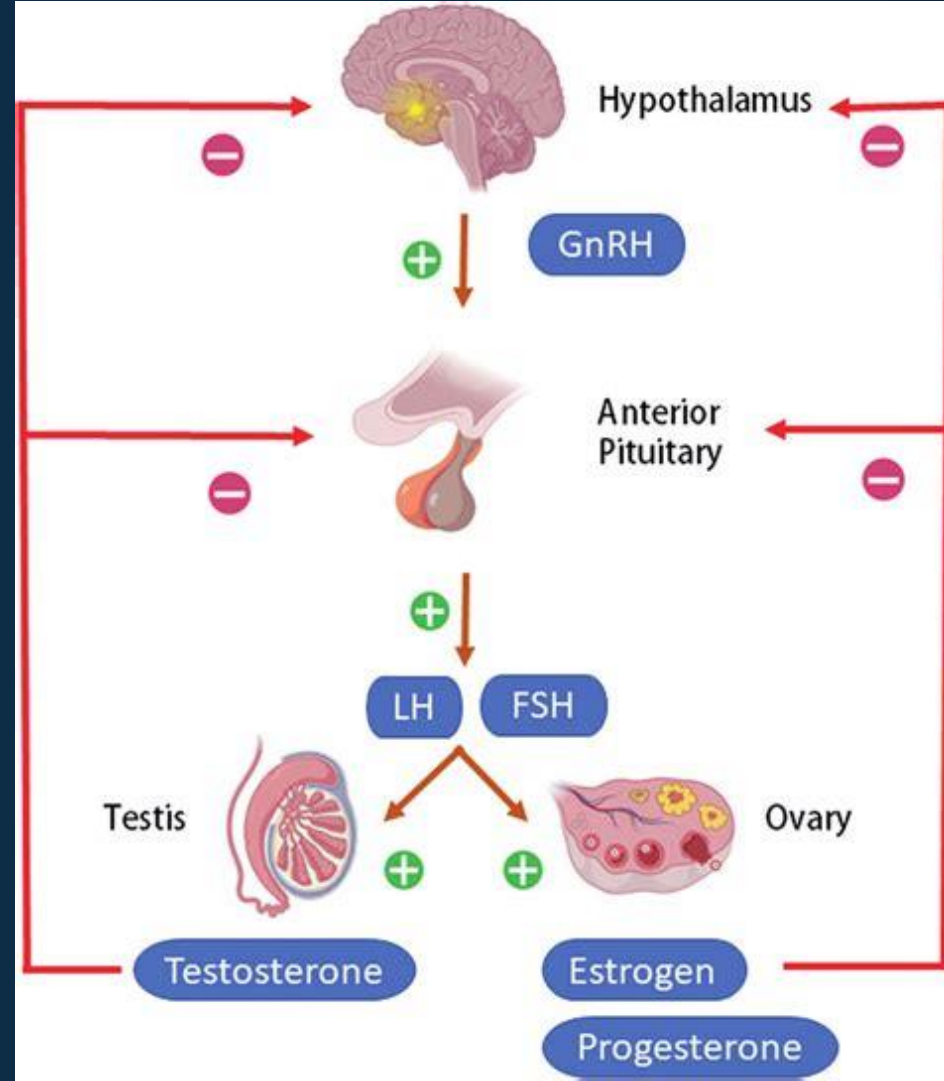
Fimbriae on Fallopian tube capture Oocyte

Oocyte travels down until ampulla (widest region of fallopian tube), waits for fertilization



Female Reproductive System





Menstrual cycle

